PMP® Demo Exam 09

Based on the *PMBOK® Guide – Eighth Edition* and the PMP® Examination Content Outline (July 2026)

Item	Real exam	This demo exam
Questions	180 (170 scored + 10 unscored pretest)	180 (all scored)
Time	240 minutes	240 minutes
Breaks	Two optional 10-minute breaks	After Q60 and after Q120
Domains	People 33% · Process 41% · Business Environment 26%	Same weighting
Approaches	~40% predictive, ~60% agile/hybrid	Same mix

How to take this exam: set a timer for 240 minutes, sit somewhere quiet, and answer every question without notes. Write your answers on paper (question number + letter). There is no penalty for guessing, so never leave a question blank. Pace yourself at about **80 seconds per question** — you should be near Q60 at 80 minutes and Q120 at 160 minutes. Only after finishing, check the Answer Key and read the explanations, then complete the score analysis.

Question types: most questions have one best answer. Multiple-response questions say how many options to choose. Matching questions ask you to pair items with options. Fill-in-the-blank questions need a number. Case-study sets share one scenario across several questions.

On screen: the same exam runs in the PMP Exam Simulator (`04_PMP_Exam_Simulator.html`), which times it, scores it by domain and task, and then gives you a PDF with the correct answer and the explanation for every question you missed.

Section 1 — Questions 1 to 60

Question 1. Which one-page tool helps a diverse group agree on a project's purpose, scope, stakeholders and main risks at the start?

- **A.** A burndown chart
- **B.** A project canvas
- **C.** A Gantt chart
- **D.** A control chart

Question 2. Which document defines the inspections and tests required at each stage of construction, and who must witness them?

- **A.** An inspection and test plan
- **B.** A stakeholder engagement plan
- **C.** A resource breakdown structure
- **D.** A procurement statement of work

Question 3. The project plan covers building and testing the system, but not training, communication or support for its users. What should the project manager do?

- **A.** Integrate change management activities into the plan
- **B.** Plan the user activities only if problems appear after go-live
- **C.** Ask the vendor to add a training course after go-live
- **D.** Leave user preparation to the line managers after go-live

Question 4. A team of 12 people must split into two teams. How should the split be decided?

- **A.** Split the team alphabetically, so that the decision is completely neutral
- **B.** Split the team by specialism, putting all of the testers together in one of the two teams
- **C.** Let the manager assign the people by seniority, so that each team has a leader
- **D.** Let the team decide how to split, keeping the skills balanced across both teams

Question 5. In an industry consortium, a competitor proposes that members share their future pricing plans. What should the project manager do?

- **A.** Share the plans, since the consortium agreement covers all information exchanges
- **B.** Decline and consult legal, since sharing such information may break competition law
- **C.** Share only approximate prices, so that no exact figures are disclosed to competitors
- **D.** Ask the other members whether they are willing to share their plans first

Question 6. The security programme is ending, but its controls must keep working. What should the closure include?

- **A.** Transferring each control to a named operational owner, with procedures and metrics
- **B.** Documenting the controls in the final report and leaving them as they are
- **C.** Switching off the controls that do not yet have an owner in the organisation
- **D.** Ask the programme team to keep running the controls informally after the programme has closed

Question 7. How many users are often enough to find most of the usability problems in one round of testing?

- **A.** Exactly one
- **B.** At least fifty
- **C.** About five
- **D.** Every user of the product

Question 8. After multi-factor authentication is introduced, employees complain that logging in takes too long. What should the project manager do?

- **A.** Acknowledge the concern, explain why, and reduce the friction where possible
- **B.** Tell the employees that security rules are not open to discussion at all
- **C.** Ignore the complaints, since employees will get used to the change over time
- **D.** Remove multi-factor authentication, since employees' satisfaction matters most

Question 9. Only two security analysts know how to respond to certain types of alert. What should the team do?

- **A.** Ask the two analysts to be on call around the clock until more analysts are hired
- **B.** Wait until one of the two analysts leaves before documenting the knowledge they have
- **C.** Write and maintain response playbooks, and rotate analysts through incidents in pairs
- **D.** Route all of those alerts to the two analysts, so that the response stays consistent

Question 10. The customer is losing confidence because minor defects keep reappearing after they have been fixed. What should the project manager do?

- **A.** Fix each defect again as quickly as possible whenever the customer reports it
- **B.** Offer the customer a discount to compensate for the recurring defects
- **C.** Explain that minor defects are normal in any new system and do not affect the system's overall value
- **D.** Find the root cause of the recurring defects, fix the process and share the improvement plan

Question 11. In the middle of a sprint, the team learns of a regulatory change that must be implemented next month. What should happen?

- **A.** The team stops the current sprint immediately and implements the change right away
- **B.** The scrum master assigns the change to the most experienced developer at once
- **C.** The team ignores the change until the regulator asks whether it has been made, to avoid rework
- **D.** The product owner adds it to the product backlog and orders it so that it is done in time

Question 12. The project manager and the vendor agree a change to a delivery date in a phone call. What should the project manager do?

- **A.** Confirm the agreement in writing through the agreed channel
- **B.** Rely on the phone call, since both parties agreed to the change
- **C.** Mention the change at the next monthly review meeting with the vendor
- **D.** Wait for the vendor to send a written confirmation, if it chooses to

Question 13. After go-live, the project team supports users intensively for four weeks before handing over to the service desk. What is this period called?

- **A.** Warranty
- **B.** Fast tracking
- **C.** Rolling wave
- **D.** Hypercare

Question 14. How can the value of a safety upgrade be expressed in the business case?

- **A.** It cannot be expressed, since safety improvements have no measurable value
- **B.** As reduced risk and avoided losses, plus compliance and reputational benefits
- **C.** As the number of safety devices installed by the end of the project
- **D.** As additional revenue from the higher production that the upgrade will allow

Question 15. Reports say that testing is 90% complete, but users report many failures in the test environment. What should the project manager do?

- **A.** Inspect the actual test results and deliverables to verify the status
- **B.** Accept the reported status, since the test lead prepared the report
- **C.** Ask the users to stop reporting failures until testing is complete
- **D.** Increase the reported figure to 95% to recognise the team's effort during testing

Question 16. Two departments with equal power must quickly reach a temporary agreement on sharing a laboratory. Full collaboration is not possible at the moment. Which approach fits best?

- **A.** Forcing
- **B.** Compromising
- **C.** Withdrawing
- **D.** Smoothing

Question 17. The client expects the vendor to absorb the cost of client-requested changes because the contract is fixed price. What should the project manager do?

- **A.** Ignore the issue until the final invoice is sent to the client
- **B.** Clarify that client-requested changes go through the contract's change process
- **C.** Refuse all changes, since the contract price cannot be adjusted
- **D.** Absorb the costs, since a fixed-price contract covers every change the client may request

Question 18. To test demand, the team adds a button for a feature that does not yet exist and measures how many users click it. What is this technique called?

- **A.** A regression test
- **B.** A fake door test
- **C.** A smoke test
- **D.** A load test

Question 19. Engagement is low after a long period of maintenance work. Which activity could renew the team's energy and creativity?

- **A.** A rule that maintenance work must be done in the evenings
- **B.** A short, timeboxed hackathon on ideas chosen by the team

- **C.** A longer sprint, so that the maintenance work is finished sooner
- **D.** A performance review for each member of the team

Question 20. The team selects a cloud platform for crew messaging. Which contract provision protects the airline if it later wants to change provider?

- **A.** A discount for paying three years in advance
- **B.** Automatic renewal of the contract
- **C.** Exit and data portability provisions
- **D.** A long minimum contract term

Question 21. Under the vendor contract, the final payment is due on acceptance after the parallel run. What should be confirmed first?

- **A.** That the acceptance criteria have been met
- **B.** That the vendor has sent its final invoice for the work
- **C.** That the users say they like the new system
- **D.** That the project budget still has enough money left

Question 22. The university buys laptops several times a year. Which procurement arrangement saves effort while remaining compliant?

- **A.** A new open tender every time a laptop is needed
- **B.** A framework agreement with pre-selected suppliers
- **C.** A single purchase of all the laptops needed for ten years
- **D.** Direct purchases from any shop without competition

Question 23. Match each change model with its focus.

(Matching: pair every numbered item with one lettered option.)

Item	Option
1. ADKAR	**A.** The stages an individual goes through to change
2. Kotter's eight steps	**B.** Endings, the neutral zone and new beginnings
3. Lewin's three stages	**C.** The steps leaders take to drive change in an organisation
4. Bridges' transition model	**D.** Unfreezing, changing and refreezing behaviour

Question 24. The project has spent 70% of its budget, and 50% of the work is complete. What does this indicate?

- **A.** The project is ahead of schedule, since it has already spent more than it planned to
- **B.** The CPI is about 1.40, which shows good cost performance
- **C.** Cost performance is poor, with a CPI of about 0.71, so an overrun is likely
- **D.** The project is on track, since about half of the work has been done

Question 25. Before a public meeting about the plant's safety, the project manager prepares three key messages, with supporting facts, for the questions that are likely to be asked. What is this technique?

- **A.** Monte Carlo analysis
- **B.** Message mapping

- **C.** Earned value analysis
- **D.** Rolling wave planning

Question 26. Lecturers upload chapters of copyrighted textbooks to the new learning platform without permission. What should the project do?

- **A.** Allow the uploads, since educational use is always exempt from copyright law
- **B.** Ask the students to report any copyrighted material that they find
- **C.** Agree a copyright and licence check with the library before publishing
- **D.** Block all uploads to the platform until the end of the academic year

Question 27. During a suspected data breach investigation, how should information be shared?

- **A.** Through the media first, so that the organisation is seen as open and transparent
- **B.** On a need-to-know basis, coordinated with legal and communications
- **C.** With all staff immediately, so that everyone can help look for the cause of the breach
- **D.** Only with the IT team, so that no one else in the organisation knows about the breach

Question 28. A contractor repeatedly breaches safety rules despite formal warnings. What may the buyer do under most contracts?

- **A.** Withhold all payments without any notice to the contractor
- **B.** Terminate for cause, following the contract's notice procedure
- **C.** Ask other contractors to do the work without informing this one
- **D.** Nothing, since safety breaches are not contractual matters

Question 29. The new online exam platform failed during an exam, and students are angry. What should the project manager do?

- **A.** Ask the faculty to deal with the students, since exams are their responsibility
- **B.** Wait until the investigation is complete before making any statement to the students
- **C.** Communicate quickly and honestly, agree fair remedies and investigate the cause
- **D.** Tell the students that the failure was caused by the large number of users logging in

Case Study — Skylane crew scheduling replacement

Skylane Airlines is replacing its 20-year-old crew scheduling system. A vendor configures a commercial product under a fixed-price contract and a predictive plan, while two internal agile teams build the integrations and a mobile app for crew members. Go-live is planned for November, the airline's quietest month.

- A trial migration of the records of **1,800 crew members** found errors in **6%** of the records.
- The vendor's work will be accepted after a **four-week parallel run**, during which the old and new systems produce rosters side by side.
- The crew schedulers, the system's main users, are needed for testing but are fully occupied running daily operations.
- Vendor work status: BAC **2, 000, 000 * *,*PV* * *1,400,000**, EV **1, 200, 000 * *,*AC* * *1,500,000**.

Questions 30–33 refer to this case.

Question 30. *(Case study: Skylane crew scheduling replacement)* What should be done about the errors found in the trial migration?

- **A.** Migrate the data as it is and let the crew schedulers correct the records after go-live
- **B.** Ask the vendor to write a script that deletes all of the records that contain errors
- **C.** Analyse the error types, fix the data at source with its owners and repeat the trial
- **D.** Accept the 6% error rate, since most of the crew records were migrated correctly

Question 31. *(Case study: Skylane crew scheduling replacement)* How should the project obtain the crew schedulers' time for testing?

- **A.** Replace the schedulers with external testers who do not know the crew rules
- **B.** Ask the schedulers to test in the evenings after their normal operational shifts
- **C.** Agree with operations a rota and backfill that release schedulers at planned times
- **D.** Skip user testing entirely, since the parallel run will find any remaining problems anyway

Question 32. *(Case study: Skylane crew scheduling replacement)* What does the to-complete performance index for the vendor's work indicate?

- **A.** A TCPI of 1.25, which is challenging but should be achievable with some extra effort from the vendor
- **B.** A TCPI of 1.00, which means the vendor's work is exactly on budget at present
- **C.** A TCPI of 0.63, which means the remaining work can easily be done within budget
- **D.** A TCPI of 1.60, which is unrealistic, so a revised estimate at completion is needed

Question 33. *(Case study: Skylane crew scheduling replacement)* What should determine whether the new system goes live after the parallel run?

- **A.** Pre-agreed go/no-go criteria, such as roster accuracy and rule compliance
- **B.** The number of defects found during the parallel run, whatever their severity
- **C.** The planned go-live date, since November is the quietest month for the airline
- **D.** The vendor's recommendation, since the vendor knows its own product best

Question 34. Business units ignore the security programme and see it as an obstacle. Which engagement approach would help most?

- **A.** Appoint security champions in each unit and involve them in prioritising the work
- **B.** Ask the chief executive to order the business units to cooperate fully
- **C.** Implement the controls without involving the business units at all
- **D.** Send a monthly email reminding the business units of the security policy and its rules

Question 35. A story covers importing crew data from five different source systems. How could it be split?

- **A.** By technical layer, starting with the database work
- **B.** It cannot be split, since all imports belong together
- **C.** By team member, giving each one a piece to build
- **D.** By source system, delivering one import at a time

Question 36. An issue that was marked as resolved has reappeared twice. What should the project manager do?

- **A.** Reopen it, analyse the root cause and verify the fix before closing it again
- **B.** Close it again, since it was resolved successfully on both previous occasions
- **C.** Create a new issue each time it reappears, so that the log stays accurate
- **D.** Remove it from the log, since recurring issues distort the project's statistics

Question 37. Aviation rules limit crew duty and flight times. How should the new crew scheduling system address these rules?

- **A.** Let the schedulers check the rules manually, since the system only proposes the rosters
- **B.** Implement the rules after go-live, once the system has proved that it works in practice
- **C.** Implement them as requirements and test them against the regulations before go-live
- **D.** Ask the vendor to confirm that its standard product already meets the regulations

Question 38. The software-as-a-service vendor releases updates every month, and each update could break the airline's integrations. What should the team do?

- **A.** Keep automated regression tests for the integrations and run them on each release
- **B.** Test the integrations manually once a year, during the airline's quietest month
- **C.** Wait for users to report problems after each update and fix them as they arise
- **D.** Ask the vendor to stop releasing updates until the airline's project has been completed

Question 39. During the weekend cutover to the new crew scheduling system, many teams must coordinate in real time. What should the project manager set up?

- **A.** A meeting on Monday morning to review what happened during the cutover
- **B.** Separate chat groups for each team, without any central coordination
- **C.** A cutover command centre with a communication schedule and clear escalation
- **D.** A shared mailbox that each team checks whenever it has time during the weekend

Question 40. Several product owners compete for the same platform team's capacity. Which governance mechanism helps most?

- **A.** A forum that agrees priorities against shared objectives, with a final say
- **B.** A rule that the platform team works on whichever request arrived first, whatever its value
- **C.** Letting the platform team choose the requests that it finds most interesting to work on
- **D.** Splitting the platform team equally between the product owners, whatever their needs

Question 41. A security risk cannot be fully mitigated before a release. Who should formally accept the residual risk?

- **A.** The tester who found the vulnerability during the security testing
- **B.** The developer who wrote the code in which the vulnerability was found
- **C.** The scrum master, since the scrum master is responsible for the process
- **D.** The risk owner with the authority to accept it, as governance defines

Question 42. New visa rules are expected to reduce international student enrolments by 20%. What should the project manager do?

- **A.** Cancel the programme immediately, since enrolments will fall

- **B.** Ask the marketing department to recruit more domestic students
- **C.** Continue as planned, since the visa rules are a matter for the government
- **D.** Reassess the business case and adjust scope or phasing with the sponsor

Question 43. Before maintenance work on a chemical line, what must be in place?

- **A.** An email from the project manager approving the work
- **B.** A verbal agreement between the operator and the fitter at the start of the shift
- **C.** A permit to work, with isolation and gas testing confirmed
- **D.** A note in the shift log that the work will take place

Question 44. The project manager realises that she tends to dominate meetings when she is under stress. Which component of emotional intelligence is she demonstrating?

- **A.** Social skills
- **B.** Motivation
- **C.** Self-awareness
- **D.** Empathy

Question 45. How can a work breakdown structure be organised?

- **A.** Only by the organisation's departments, so that each department owns a branch
- **B.** Only alphabetically, so that work packages are easy to find
- **C.** Only by time periods, with one branch for each month of the project
- **D.** By major deliverables, by phases or by subprojects, depending on the project

Question 46. Students expect new features during the exam period, but the team has planned a change freeze then. What should the product owner do?

- **A.** Communicate the freeze and its reasons in advance, and show when the features will come
- **B.** Release the features secretly during the exams and fix any problems as they appear
- **C.** Cancel the freeze and release the new features during the exams as the students expect
- **D.** Say nothing about the freeze, since students do not need to know about the team's release planning

Question 47. When is an agile release plan typically updated?

- **A.** Only at the start of the release, since the plan is a commitment
- **B.** Once a quarter, at the same time as the financial forecasts
- **C.** Every sprint, based on the latest velocity and backlog changes
- **D.** Only when the sponsor formally requests a change to the plan

Question 48. Which two factors are typically used to prioritise issues in the issue log? (Choose two.)

- **A.** The urgency of resolving the issue
- **B.** The impact on the project's objectives
- **C.** The date on which the issue was raised
- **D.** The alphabetical order of issue titles
- **E.** The seniority of the person who raised it

Question 49. Which two actions help a leader guide a team through a long period of uncertainty? (Choose two.)

- **A.** Sharing what is known and unknown, honestly and often
- **B.** Focusing the team on what it can control
- **C.** Waiting until everything is certain before communicating
- **D.** Promising outcomes that cannot be guaranteed
- **E.** Avoiding discussion of the team's concerns

Question 50. Key engineers in a three-year programme are being approached by other companies. What should the project manager do?

- **A.** Ask human resources to add penalty clauses to the engineers' contracts
- **B.** Discuss their career goals and offer development opportunities and recognition
- **C.** Assign them less interesting work so that they are less visible to recruiters
- **D.** Do nothing, since engineers always change jobs every few years anyway

Question 51. The team must document a complex student appeals process with many alternative paths and exceptions. Which technique fits best?

- **A.** A list of the appeals office's staff
- **B.** A single user story for the whole process
- **C.** A Gantt chart of the appeals process
- **D.** Use cases with main and alternative flows

Question 52. Training on the new procedures is scheduled only on weekday mornings, but the operators work rotating shifts. What should the project do?

- **A.** Train only the day-shift operators and let them teach the others
- **B.** Send the night-shift operators the slides to read at home
- **C.** Ask the night-shift operators to come in on their days off
- **D.** Schedule training for every shift, including nights and weekends

Question 53. Several key academic approvers are on sabbatical, and their sign-offs are delayed. What should the project manager do?

- **A.** Ask the approvers' students to sign on their behalf
- **B.** Wait until the approvers return from their sabbaticals
- **C.** Proceed without the sign-offs and inform the approvers on their return
- **D.** Agree delegates with authority through the faculty's governance

Question 54. During the shutdown, 600 contract workers will be on site. What must the resource plan include?

- **A.** Only the number of workers needed for each day of the shutdown period, by trade and shift
- **B.** Mobilisation logistics, such as inductions, accommodation, permits and supervision
- **C.** Only the names of the contractors that will supply the workers
- **D.** Nothing extra, since the contractors manage their own workers on site

Question 55. The estimate at completion is $750,000, and the actual cost to date is $300,000. What is the estimate to complete? (Answer in dollars.)

(Fill in the blank.)

Answer: _____

Question 56. The vendor's team complains that the buyer's product owner takes weeks to answer questions and approve stories. What should the project manager do?

- **A.** Tell the vendor to proceed without approvals and correct any mistakes later
- **B.** Agree response times for decisions and secure the product owner's availability
- **C.** Ask the vendor to send its questions in writing once a month in a single list
- **D.** Replace the vendor with one that needs fewer decisions from the product owner

Question 57. Operators at the plant rarely report near misses because they fear being blamed. What should the project manager do?

- **A.** Offer a cash reward for every near miss that an operator reports to management
- **B.** Make near-miss reporting compulsory and discipline operators who fail to report them
- **C.** Promote a just culture: thank reporters, focus on systems and act on reports
- **D.** Stop collecting near-miss reports, since operators do not want to report them

Question 58. The project credits 50% of an activity's value when the activity starts and the remaining 50% when it finishes. What is this earned value method called?

- **A.** The 50/50 rule
- **B.** Apportioned effort
- **C.** Weighted milestones
- **D.** The 0/100 rule

Question 59. Researchers object to new security controls that block software they use in their experiments. What should the project manager do?

- **A.** Enforce the controls without exceptions, since security policies apply equally to all university staff
- **B.** Facilitate a discussion to find solutions meeting both needs, such as approved exceptions
- **C.** Remove the controls for the research departments, since academic freedom is more important than security
- **D.** Ask the researchers to use their own personal laptops for the experiments that need the blocked software

Question 60. Who should accept the new crew scheduling system into operations?

- **A.** The project manager, once the project's testing has been completed
- **B.** The vendor, since it built and configured the scheduling system
- **C.** The operational owner, against agreed operational acceptance criteria
- **D.** The internal audit department, after an audit of the project

> ☕ **Optional 10-minute break.** In the real exam, once you start a break you cannot return to the questions of the previous section, so review flagged questions before you go.

Section 2 — Questions 61 to 120

Question 61. The shutdown requires welders certified for work on pressure equipment. What should the project manager make sure of?

- **A.** That the contractor confirms verbally that its welders are qualified
- **B.** That the plant's own staff supervise welders who lack certification
- **C.** That the welders are certified within a month after the shutdown has finished
- **D.** That the certifications are verified before the workers are mobilised

Question 62. A board member keeps asking for detailed technical information, which the other board members find distracting. What should the project manager do?

- **A.** Stop answering the member's questions in meetings, since the other board members find them distracting
- **B.** Include full technical detail in every board update, so that this member is satisfied
- **C.** Offer that member a separate technical briefing, and keep board updates at the right level
- **D.** Ask the chair to tell the member to stop asking technical questions in the board meetings

Question 63. Team members disagree about whether cameras must be switched on in video meetings. What should the scrum master do?

- **A.** Require cameras to be on in every meeting from now on
- **B.** Ask the functional managers to decide the rule for the team
- **C.** Help the team discuss its needs and agree a working agreement
- **D.** Let each person do whatever they prefer, without any discussion in the team

Question 64. What size does Scrum recommend for a Scrum team?

- **A.** Typically 10 or fewer people
- **B.** At least 15 people
- **C.** Any size, as long as there is one scrum master
- **D.** Exactly 7 developers

Question 65. In a negotiation, the vendor opens with a very high price to influence the final outcome. What is this tactic called?

- **A.** Fait accompli
- **B.** Good guy, bad guy
- **C.** Smoothing
- **D.** Anchoring

Question 66. A chemical plant upgrade could change the emergency planning arrangements for the surrounding area. Which stakeholders must be engaged?

- **A.** Local emergency services and the authorities responsible for emergency planning
- **B.** Only the plant's own employees, since they work with the hazardous materials
- **C.** The plant's customers, since they buy the products made at the plant
- **D.** Only the plant's shareholders, since they own the plant and are liable for its hazards

Question 67. The team deploys to production several times a day but plans its work in two-week sprints. Is this consistent with Scrum?

- **A.** No; a team that deploys continuously must use Kanban instead of Scrum
- **B.** Yes; increments can be released at any time, and the sprint is a planning cadence
- **C.** Yes, but only if the product owner approves each deployment in writing
- **D.** No; Scrum allows a release only once, at the end of each sprint after the sprint review

Question 68. Before inviting contractors to bid for shutdown work, what should the plant check?

- **A.** Only whether they are available on the shutdown dates
- **B.** Only their prices from previous contracts with the plant, to find the cheapest
- **C.** Nothing, since the bids will show which contractor is best
- **D.** Their safety performance and qualifications, through pre-qualification

Question 69. A new research computing service will serve all faculties. Who should decide its priorities after launch?

- **A.** The vendor that supplies the service's hardware and software
- **B.** The project team, which will stay in place after the launch
- **C.** The faculty that shouts loudest about its needs each month
- **D.** A service owner, supported by a board representing the faculties

Question 70. A serious accident has occurred at another company's similar plant. How should the project use this?

- **A.** Review the published findings and apply the relevant lessons to the design and procedures
- **B.** Use it only in the project's public communications to show that the plant is safer
- **C.** Wait for the regulator to issue new rules before considering any of the lessons from the accident
- **D.** Ignore it, since the accident happened at another company with different staff

Question 71. A security incident reveals a type of vulnerability that affects many features. How should the work be handled?

- **A.** The product owner reprioritises the backlog, using the agreed expedite route
- **B.** The team stops all work until every feature has been rewritten from the beginning
- **C.** The team finishes the current sprint as planned and adds the fixes at the end of the release
- **D.** The security team fixes the code itself without telling the product owner or the developers

Question 72. New regulations will restrict how the plant can transport chlorine to its customers. What should the project manager do?

- **A.** Ask the customers to collect the chlorine themselves so that the rules do not apply
- **B.** Assess the impact on operations and the project with logistics and compliance
- **C.** Wait until the new transport rules are in force before assessing what they mean
- **D.** Continue as planned, since the project upgrades the plant and not the transport arrangements

Question 73. Which mechanism ensures that work starts only when it has been formally approved, at the right time and in the right sequence?

- **A.** A lessons learned register
- **B.** A stakeholder register

- **C.** A product roadmap
- **D.** A work authorisation system

Question 74. During a ransomware incident, many people give conflicting instructions to the response team. What should the leader do first?

- **A.** Ask the most senior executive present to take over all technical decisions personally
- **B.** Let everyone continue to act on their own judgement, since speed matters most in an incident
- **C.** Wait for the external forensic consultants to arrive before giving any instructions
- **D.** Establish clear command: one incident lead, defined roles and a regular update rhythm

Question 75. Users expect all 20 years of historical records to be migrated to the new system, but the plan migrates five years and archives the rest. What should the project manager do?

- **A.** Say nothing until go-live, since most users will not need the older records anyway
- **B.** Delete the older records, since they will no longer be needed in the new system
- **C.** Explain the approach and how archived data can be accessed, and agree it with the owner
- **D.** Migrate all 20 years of data to avoid disappointing the users, whatever it costs the project

Question 76. The team has spent 60% of the release budget but delivered only 40% of the planned scope. What does this suggest?

- **A.** The release is likely to overrun its budget unless the scope or the plan changes
- **B.** The release is on track, since spending usually runs ahead of delivery in agile
- **C.** The budget should be increased automatically by 50% to match the spending
- **D.** The team should stop measuring scope, since agile teams do not track scope against a budget

Question 77. Project managers say that the PMO's templates are slow and difficult to use. What should the PMO do?

- **A.** Make the templates mandatory and audit their use more often in every project
- **B.** Ask the project managers to attend training on the templates
- **C.** Remove all templates and let each project work as it wishes
- **D.** Gather feedback, simplify the templates and measure the effect

Question 78. Which two of the following are intangible benefits? (Choose two.)

- **A.** A stronger brand reputation
- **B.** Lower maintenance costs
- **C.** Improved staff morale
- **D.** Reduced energy costs
- **E.** Higher sales revenue

Question 79. In what order are costs aggregated to form the cost baseline?

- **A.** Activities, work packages, control accounts, project
- **B.** Work packages, activities, project, control accounts
- **C.** Project, control accounts, work packages, activities
- **D.** Control accounts, project, activities, work packages

Question 80. Team members sign up for a free online tool to share files with a vendor, outside the approved IT channels. What should the scrum master do?

- **A.** Ask the vendor to choose a file-sharing tool that suits it best
- **B.** Allow it for non-confidential files only, without telling the IT department
- **C.** Allow it, since the tool is free and makes collaboration with the vendor easier
- **D.** Switch to an approved tool, or request approval through the proper process

Question 81. Which activity tests a system's security by simulating real attacks?

- **A.** Unit testing
- **B.** Penetration testing
- **C.** Load testing
- **D.** Usability testing

Question 82. How should schedule baselines be handled in a hybrid project?

- **A.** Baseline only the agile work, since the predictive work is managed by the vendors
- **B.** Baseline every user story in the agile backlog, so that all of the work is controlled in the same way
- **C.** Avoid baselines entirely, since a hybrid project cannot use any predictive techniques
- **D.** Baseline the predictive work, forecast the agile work from velocity and link them through milestones

Question 83. With limited time, how should the team decide where to focus its testing?

- **A.** On the areas that the developers changed least during the last sprint
- **B.** On the areas that are easiest to test, so that more of the tests can be completed in time
- **C.** On the areas with the highest risk, based on the likelihood and impact of failure
- **D.** Equally on every area, so that no part of the product is treated differently

Question 84. A powerful department head blocks decisions in group meetings without explaining why. What should the project manager do first?

- **A.** Meet her one-to-one to understand her concerns
- **B.** Exclude her from future decision meetings
- **C.** Escalate her behaviour to the sponsor immediately
- **D.** Make decisions when she is not present

Question 85. Critical vulnerabilities must be fixed within 72 hours, while other work follows the normal flow. Which Kanban practice supports this?

- **A.** An expedite class of service with its own policy
- **B.** A longer sprint for vulnerability work, so that all fixes fit
- **C.** A separate team for every vulnerability
- **D.** Removing the WIP limits for all work

Question 86. How can a chemical plant make sure that staff know how to act in an emergency that rarely happens?

- **A.** An annual email with the emergency phone numbers
- **B.** A poster in the canteen describing the procedures

- **C.** A copy of the emergency plan for each employee
- **D.** Regular drills and tabletop exercises

Question 87. Which response best addresses the risk of losing student records?

- **A.** A rule that staff must be careful with the records
- **B.** Storing all records on one very reliable server
- **C.** Regular backups with tested recovery procedures
- **D.** An insurance policy that covers lost student records

Question 88. At the end of the final sprint, two stories are unfinished. What happens to them?

- **A.** They are not done, and they go back to the product backlog for the product owner
- **B.** They are counted as done, since most of the work on them has been completed by the team
- **C.** They are deleted, since the project has ended and nobody will work on them any more
- **D.** They are moved into the sprint report as completed work with a note explaining the gaps

Question 89. Three risks have the following probabilities and impacts: 10% and $200,000; 30% and $50,000; 5% and $1,000,000. What contingency reserve do their expected monetary values suggest? (Answer in dollars.)

(Fill in the blank.)

Answer: _____

Question 90. With a limited budget, how should the security team order its backlog of possible controls?

- **A.** By implementing the most expensive controls first, since they are best
- **B.** By the risk reduction each control achieves relative to its cost
- **C.** By how interesting each control is for the security team to build
- **D.** By the order in which the controls were suggested by the auditors

Question 91. The team deliberately causes failures in a controlled way to find weaknesses in its systems before real incidents do. What is this practice called?

- **A.** Fast tracking
- **B.** Gold plating
- **C.** Chaos engineering
- **D.** Regression testing

Question 92. Management sets a target of 100% code coverage, and developers start writing tests that check nothing. What should the team do?

- **A.** Ask the testers to write all of the tests so that developers can code
- **B.** Focus on meaningful tests and use coverage as an indicator, not a target
- **C.** Stop measuring code coverage, since all quality metrics are misleading in practice
- **D.** Keep the target, since 100% coverage proves that the code has no defects

Question 93. A feature will earn $20,000 per week once it is released. If its release is delayed by three weeks, what is the cost of the delay?

- **A.** $20,000
- **B.** $40,000

- **C.** $60,000
- **D.** $80,000

Question 94. Employees keep clicking on phishing emails. Which approach is most likely to change their behaviour over time?

- **A.** Publicly naming the employees who click on phishing emails
- **B.** A single long training session once a year for all staff
- **C.** Regular, realistic simulations with immediate, supportive training
- **D.** Block all external emails from reaching employees' inboxes until behaviour improves

Question 95. Cabin crew, the main users of a new app, work irregular shifts and are rarely in the office. How should the team engage them?

- **A.** Invite them to sprint reviews held at the head office on weekdays
- **B.** Use short mobile surveys, in-app feedback and sessions at crew bases
- **C.** Ask their managers to describe what the cabin crew need from the app
- **D.** Wait until the app is launched and then collect feedback from the crew

Question 96. A critical feature depends on an open-source library maintained by a single volunteer. What should the team do?

- **A.** Ignore the risk, since open-source libraries are always maintained by the community
- **B.** Remove the feature from the product immediately to avoid any dependency
- **C.** Assess the risk and plan responses, such as preparing an alternative
- **D.** Copy the library's code into the product and stop tracking its updates

Question 97. An activity has an optimistic estimate of 4 days and a pessimistic estimate of 22 days. What is its PERT standard deviation, in days? (Answer with a number.)

(Fill in the blank.)

Answer: _____

Question 98. A team describes where it wants its process to be in three months and works towards it in small experiments, step by step. What is this approach called?

- **A.** Earned value management
- **B.** The critical path method
- **C.** Rolling wave planning
- **D.** The improvement kata

Question 99. The plant manager expects no production downtime, but the safety upgrade requires a ten-day shutdown. What should the project manager do?

- **A.** Explain why the shutdown is needed and plan it with operations, e.g. during scheduled maintenance
- **B.** Postpone the upgrade indefinitely until the plant manager accepts some downtime for the project
- **C.** Ask the sponsor to overrule the plant manager, since safety upgrades take priority over production
- **D.** Plan the upgrade without a shutdown, even if this means carrying out the work while the plant is running

Question 100. The sprint burndown stays flat for most of the sprint and then drops sharply on the last day. What does this suggest?

- **A.** Stories are finished only at the end; the team should split work and finish items steadily
- **B.** The team is working very efficiently, since all of the work is completed within the sprint
- **C.** The product owner is adding new work each day, which keeps the burndown chart flat
- **D.** The burndown chart is broken and should be replaced with a different tool for the team

Question 101. The board expects the new security programme to make the company completely immune to cyberattacks. What should the project manager do?

- **A.** Explain that risk can be reduced but not eliminated, and agree measurable targets
- **B.** Avoid the subject, since the board members are not experts in cybersecurity
- **C.** Promise immunity for the first year only, while the programme is still running
- **D.** Confirm the board's expectation, since the programme has a large budget and expert staff

Question 102. A chemical plant has a 21-day maintenance shutdown every four years. How should it be planned?

- **A.** Day by day during the shutdown, since the work found inside the equipment cannot be predicted
- **B.** Months in advance, with an early scope freeze and a detailed shutdown schedule
- **C.** In the last few weeks before the shutdown, once the condition of the equipment is known
- **D.** In the same way as normal daily maintenance, since the same maintenance team will do the work

Question 103. A shift cannot end until the next shift has started, so that handover can take place. Which type of dependency is this?

- **A.** Start-to-finish
- **B.** Start-to-start
- **C.** Finish-to-start
- **D.** Finish-to-finish

Question 104. The organisation is moving from temporary project teams to long-lived product teams. Which change is most important to support this?

- **A.** Renaming the project managers as product managers
- **B.** Funding persistent teams and defining product ownership roles
- **C.** Asking each team to choose a new name and logo
- **D.** Keeping the annual project funding process as it is

Question 105. Operators at a chemical plant see the new process safety programme as extra paperwork. What would best build a shared sense of purpose?

- **A.** Remove some of the programme's checks so that the operators see less paperwork in their daily work
- **B.** Connect the programme to real incidents and to everyone going home safely
- **C.** Explain that the programme is mandatory under corporate policy and will be audited every quarter
- **D.** Offer a bonus to the operators who complete the most safety forms each month

Question 106. The company outsources 24/7 security monitoring to a managed security provider. What should the contract define?

- **A.** Service levels, such as detection and response times, with reporting and remedies
- **B.** Nothing specific, since a long-term relationship is based on trust
- **C.** Only the number of analysts that the provider must employ
- **D.** Only the monthly fee, since the provider is the expert and knows how to monitor security

Question 107. Which two techniques help estimate when a large backlog will be completed? (Choose two.)

- **A.** Assuming that the best sprint will repeat
- **B.** Monte Carlo simulation based on throughput
- **C.** Ignoring past performance data
- **D.** Asking the sponsor for a preferred date
- **E.** Velocity-based forecasting that gives a range

Question 108. Match each contract type with its main feature.

(Matching: pair every numbered item with one lettered option.)

Item	Option
1. Firm fixed price	**A.** The price is adjusted in line with defined indexes, such as material costs
2. Cost plus award fee	**B.** The price is fixed for the defined scope, and the seller bears the cost risk
3. Time and materials	**C.** Costs are reimbursed, plus a fee based on the buyer's judgement of performance
4. Fixed price with economic price adjustment	**D.** The buyer pays agreed rates for time and materials used

Question 109. Match each motivation theory with its key idea.

(Matching: pair every numbered item with one lettered option.)

Item	Option
1. Maslow	**A.** A hierarchy of needs, from basic needs to self-actualisation
2. Herzberg	**B.** Hygiene factors prevent dissatisfaction, while motivators drive satisfaction
3. McClelland	**C.** Motivation depends on expectancy, instrumentality and valence
4. Vroom	**D.** Needs for achievement, affiliation and power

Question 110. After acceptance, the customer expects new features to be built free of charge under the warranty. What should the project manager do?

- **A.** Ask the team to build the features in their spare time without charging
- **B.** Explain that the warranty covers defects; new features are changes
- **C.** Refuse to discuss the features until the warranty period has ended
- **D.** Build the features for free to keep the customer satisfied with the project

Question 111. The product backlog has grown to 900 items, many of which are years old. What should the product owner do?

- **A.** Regularly remove or archive items that are no longer relevant
- **B.** Start a new backlog and delete the old one without reviewing any of the items
- **C.** Ask the developers to estimate all 900 items in detail
- **D.** Keep every item, since each one was requested by someone

Question 112. Critical instruments are held at customs because some shipping documents are missing. What should the project manager do?

- **A.** Order new instruments from another supplier, since the shipment may take months to release
- **B.** Wait for customs to contact the project, since customs procedures are outside its control
- **C.** Work with the supplier and customs agent to supply the documents, and assess the impact
- **D.** Ask the customs officers to release the instruments without the missing documents

Question 113. Which two inputs are most useful for sprint planning? (Choose two.)

- **A.** The risk management plan
- **B.** The project charter
- **C.** The ordered product backlog
- **D.** The lessons learned repository
- **E.** The team's capacity and past performance

Question 114. What should be the main basis for the change control board's decision on a change request?

- **A.** Whether the project team would enjoy working on the requested change
- **B.** Its impact on the project's objectives and value, compared with its costs and risks
- **C.** The seniority of the person who submitted the change request to the board
- **D.** The number of change requests that the board has already approved this year

Question 115. The cloud provider announces a 30% price increase. What should the product owner do?

- **A.** Move to another provider immediately, without assessing the effort involved
- **B.** Accept the increase, since moving to another provider is never possible
- **C.** Pass the full increase on to customers straight away, without considering the effects
- **D.** Assess the cost impact and options, such as optimisation or alternative providers

Question 116. Developers see security as the security team's job rather than their own. What would best build shared ownership?

- **A.** Make security part of the team's goals and definition of done, with experts in planning
- **B.** Keep security separate, since developers should focus on delivering new features to customers
- **C.** Send developers a monthly report showing the number of security defects found in their code
- **D.** Ask the security team to review every story at the end of each sprint and report any problems

Question 117. Each extra day of the shutdown costs $500,000 in lost production. Crashing an activity on the critical path would save two days and cost $300,000. What should the project manager recommend?

- **A.** Do not crash it, since lost production is not a project cost

- **B.** Crash it only if the contractor agrees to pay the $300,000
- **C.** Crash the activity, since it saves $1,000,000 for a cost of $300,000
- **D.** Do not crash it, since crashing always adds risk to a shutdown

Question 118. A newly formed team keeps arguing about priorities. The coach suggests a short team mission statement. How should it be created?

- **A.** By the sponsor, who knows the organisation's strategy
- **B.** By copying the mission statement of another team
- **C.** By the team together, linked to the product vision
- **D.** By the product owner alone, before the next sprint

Question 119. Users do not know what has changed in each release. What should the team do?

- **A.** Ask users to look for the changes themselves in the product
- **B.** Send users the full technical change log of every commit
- **C.** Hold a two-hour training session for all users after every release
- **D.** Publish short, user-focused release notes with each release

Question 120. How can the team capture the ways in which an attacker might misuse a new feature?

- **A.** Wait for the penetration test before release
- **B.** Add a general "make it secure" item to the backlog
- **C.** Ask users to describe how they will use the feature
- **D.** Write misuse stories alongside the user stories

> ☕ **Optional 10-minute break.** In the real exam, once you start a break you cannot return to the questions of the previous section, so review flagged questions before you go.

Section 3 — Questions 121 to 180

Question 121. Stakeholders receive conflicting information from several spreadsheets and the team's backlog tool. What should the team do?

- **A.** Update all of the spreadsheets and the backlog tool every day to keep them aligned
- **B.** Make the backlog tool the single source of truth and retire the spreadsheets
- **C.** Ask stakeholders to compare the documents and tell the team about any differences
- **D.** Send all of the documents to stakeholders, so that they can choose the one they prefer

Question 122. The plant's insurer could lower its premiums if certain safety improvements are made. What should the project manager do?

- **A.** Ask the insurer to pay for the safety improvements in exchange for lower premiums
- **B.** Wait until the upgrade is complete and then ask the insurer to review the premiums
- **C.** Engage the insurer early to understand its criteria, and include them where they add value
- **D.** Ignore the insurer, since it has no formal role in the plant's safety upgrade project

> **Case Study — Halden Chemicals safety upgrade**
>
> Halden Chemicals is upgrading the control and safety systems of a chlorine production unit at a site regulated as a major-hazard installation. Hardware and installation follow a predictive plan; an agile team builds a new digital operations dashboard.
>
> - Corporate rules require every project at a major-hazard site to pass an **independent stage-gate review** before construction starts.
> - Any change to safety systems must go through the site's **management-of-change (MOC)** process, and the regulator must be notified before restart.
> - A hazard and operability study (HAZOP) of the new design produced **40 recommendations**, **6** of them rated high-risk.
> - Control room operators will move from panel controls to screens. Experienced operators worry that they will lose their overview of the process.
>
> *Questions 123–126 refer to this case.*

Question 123. *(Case study: Halden Chemicals safety upgrade)* The contractor is ready to start construction, but the independent stage-gate review has not yet taken place. What should the project manager do?

- **A.** Ask the site manager to waive the review, since the contractor is already mobilised
- **B.** Ask the contractor to start with preparatory work that the review would not cover
- **C.** Start construction now and hold the review in parallel, so that no time is lost
- **D.** Wait for the review, as corporate governance requires, and prepare for it promptly

Question 124. *(Case study: Halden Chemicals safety upgrade)* During installation, engineers want to change a safety interlock setting to speed up commissioning. What must happen?

- **A.** The project manager approves the change verbally, since commissioning is running behind schedule
- **B.** It goes through management of change for review and approval before it is made

- **C.** The change is made temporarily and reversed after commissioning, without the need for any review
- **D.** The engineers make the change and record it in the commissioning log for later review by the team

Question 125. *(Case study: Halden Chemicals safety upgrade)* How should the six high-risk HAZOP recommendations be handled?

- **A.** Treat all 40 recommendations the same and close them in the order in which they were raised
- **B.** Assign owners, design responses and track each one to verified closure before restart
- **C.** Address them after restart, once the plant has returned to normal production
- **D.** Accept them, since the HAZOP team included experienced engineers from the plant

Question 126. *(Case study: Halden Chemicals safety upgrade)* How should the project address the operators' worry about losing their overview of the process?

- **A.** Keep the old control panels in the control room as a backup, so that operators can use them if needed
- **B.** Tell the operators that modern screens are better than panels, so their worries are unfounded
- **C.** Involve operators in designing and testing the screens, and train them on a simulator before go-live
- **D.** Train the operators on the new screens during the first week after the system goes live

Question 127. When should the team consider how attackers could misuse new features?

- **A.** Only when the security team asks for an assessment
- **B.** At the end of the project, during the final security audit
- **C.** During refinement and design, through threat modelling
- **D.** After release, when the first attacks have been detected

Question 128. A project team has 66 communication channels. How many people are on the team, including the project manager? (Answer with a number.)

(Fill in the blank.)

Answer: _____

Question 129. Steering committee meetings spend most of their time on items that are on track. What should the project manager change?

- **A.** Add more detail about the items that are on track, so that the committee is reassured
- **B.** Hold the meetings more often, so that every item can be discussed in full
- **C.** Stop reporting to the committee until a serious problem occurs
- **D.** Use exception-based reporting, focused on variances, risks and decisions needed

Question 130. The project manager classifies stakeholders by their influence over the project and the impact the project has on them. Which tool is this?

- **A.** A RACI chart
- **B.** A resource histogram
- **C.** A probability/impact matrix
- **D.** An influence/impact grid

Question 131. During a plant shutdown, how should progress be tracked?

- **A.** Only when a contractor reports that its work is running late
- **B.** At the end of the shutdown, by comparing the final cost with the budget
- **C.** Weekly, using the contractors' own estimates of their percentage complete
- **D.** Daily, using earned hours against the detailed plan, with quick corrective action

Question 132. What must happen before the plant is handed back to operations after a shutdown?

- **A.** All of the contract workers demobilised and their accommodation closed
- **B.** The final invoices from all contractors received and paid in full by the plant
- **C.** The lessons learned report approved by the sponsor and the plant's board
- **D.** Permits closed, temporary equipment removed and the pre-startup review done

Question 133. How should the airline measure the value of the new crew scheduling system?

- **A.** Through the total number of hours that the project team worked on the implementation
- **B.** Through the vendor's own satisfaction survey of the project team at the end of the project
- **C.** Through the number of screens and reports that the vendor configured in the new system
- **D.** Through outcomes such as fewer crew-related delays, compared with a baseline

Question 134. Which two practices help a distributed security team keep a common vision? (Choose two.)

- **A.** Revisiting the goals regularly in joint planning sessions
- **B.** Sharing the vision only with the team leads
- **C.** Letting each location set goals of its own
- **D.** Making goals and progress visible to every location
- **E.** Presenting the vision once, at the kickoff

Question 135. Small groups of front-line workers meet regularly to identify and solve problems in their own work. What is this practice called?

- **A.** Phase gate reviews
- **B.** Quality circles
- **C.** Change control boards
- **D.** Steering committees

Question 136. A new student system requires changes to academic regulations, which the faculty senate must approve. What should the project manager do?

- **A.** Avoid the senate by describing the changes as technical updates
- **B.** Ask the IT department to present the changes to the senate at the end
- **C.** Implement the system first and ask the senate to approve afterwards
- **D.** Engage the senate early and plan the approval into the schedule

Question 137. A developer wants to become a scrum master. How should the current scrum master support her?

- **A.** Agree a development plan, such as facilitating some events with coaching and feedback
- **B.** Suggest that she reads the Scrum Guide on her own and takes a certification exam
- **C.** Discourage her, since the team needs her skills as a developer more than as a scrum master

- **D.** Tell her to wait until a scrum master position becomes available in another team

Question 138. After go-live, staff still follow the old procedures because the official procedures were never updated. What should the project do?

- **A.** Update the procedures, roles and training as part of the change
- **B.** Tell the staff that they must follow the new system, whatever the procedures say
- **C.** Leave the procedures as they are, since the staff will adapt over time
- **D.** Ask each department to write its own procedures without any guidance

Question 139. What is the present value of $121,000 received in two years, using a discount rate of 10%?

- **A.** $100,000
- **B.** $110,000
- **C.** $121,000
- **D.** $133,100

Question 140. Academic staff on the project are often unavailable because of their teaching duties. What should the project manager do?

- **A.** Ask them to cancel some of their classes, so that they can work on the project instead
- **B.** Plan around the teaching timetable and agree their allocation with the department head
- **C.** Proceed without their input and ask them to review everything at the end
- **D.** Replace them with external consultants who have no knowledge of the university

Question 141. Testers cannot use production data because of privacy rules, and testing is blocked. What should the team do?

- **A.** Copy the production data into the test environment quietly, just for this release
- **B.** Skip the tests that need realistic data and test those features in production after the release
- **C.** Create anonymised or synthetic test data, agreed with the data protection officer
- **D.** Wait until the privacy rules change before continuing with the blocked testing

Question 142. A research grant's budget may only be spent within its approved categories. The team wants to move money from travel to equipment. What should it do?

- **A.** Request approval from the funder before moving the money
- **B.** Buy the equipment now and report the change in the final grant report
- **C.** Move the money, since the total budget does not change
- **D.** Ask the university's finance office to hide the transfer

Question 143. One bid is 45% lower than all of the others. What should the evaluation team do?

- **A.** Reject the bid immediately, since it must contain a serious mistake
- **B.** Ask the bidder to explain the price, and check that it has understood the scope
- **C.** Award the contract to this bidder at once, since it offers the best price
- **D.** Ask the other bidders to lower their prices to match this one before the evaluation is finished

Question 144. Stakeholders are upset that items labelled "should have" were not delivered in the release. What should the product owner do in future?

- **A.** Deliver all "should have" items, whatever the date impact

- **B.** Clarify what each MoSCoW category means when priorities are set
- **C.** Stop using MoSCoW and prioritise items without telling anyone
- **D.** Label every item as "must have" to avoid disappointment

Question 145. A hazard and operability study (HAZOP) recommends an additional pressure relief valve. How should this be handled?

- **A.** Add it to the requirements and design, traced back to the HAZOP recommendation
- **B.** Ignore the recommendation, since the design has already been approved
- **C.** Record it in the lessons learned register for the next project to consider
- **D.** Install the valve on site without updating the design, since it is a small change

Question 146. A remote team member has become quiet and disengaged in recent sprints. What should the scrum master do first?

- **A.** Mention the problem in the next retrospective in front of the whole team
- **B.** Report the team member's lack of engagement to the functional manager
- **C.** Wait to see whether the team member's engagement improves on its own
- **D.** Have a private one-to-one conversation to understand what is going on

Question 147. The budget covers the new system, but not training, backfilling staff while they are in training, or communications. What should the project manager do?

- **A.** Add these change management costs to the estimate
- **B.** Cover them from the management reserve without approval
- **C.** Ask the vendor to provide them free of charge
- **D.** Leave them out, since departments pay for their own staff

Question 148. In critical chain scheduling, where is the project buffer placed?

- **A.** Inside each activity on the critical chain
- **B.** Wherever a non-critical chain of activities joins the critical chain
- **C.** At the end of the critical chain, before the project finish date
- **D.** At the start of the project, before the first activity

Question 149. Which chart ranks the risks or variables that have the greatest effect on a project's outcome, from largest to smallest?

- **A.** A tornado diagram
- **B.** A Gantt chart
- **C.** A histogram
- **D.** A cause-and-effect diagram

Question 150. The team wants to connect a business goal to the changes in behaviour, and the features, that could achieve it. Which technique fits best?

- **A.** Earned value analysis
- **B.** A RACI chart
- **C.** Impact mapping
- **D.** Resource levelling

Question 151. Team members ask how their project contributes to the organisation's strategy. What should the project manager do?

- **A.** Ask the sponsor to answer their questions at the end of the project
- **B.** Tell them that strategy is a matter for senior management, not for the team
- **C.** Send them a copy of the organisation's annual report to read in their own time
- **D.** Show how the strategy links to the project's objectives and to their own work

Question 152. When should a scrum master escalate an impediment beyond the team?

- **A.** Immediately, every time a team member mentions any kind of problem
- **B.** Never, since self-managing teams must solve all of their own problems without outside help
- **C.** Only at the end of the release, when all of the impediments are known
- **D.** When the team cannot remove it and it threatens the sprint goal or delivery

Question 153. Which technique shows a hazardous event with its causes and preventive barriers on one side, and its consequences and mitigating barriers on the other?

- **A.** A burnup chart
- **B.** A fault tree analysis
- **C.** A Pareto chart
- **D.** A bow-tie analysis

Question 154. In a requirements workshop for a crew scheduling system, union representatives and airline managers argue about how certain roster rules should be interpreted. What should the project manager do?

- **A.** Choose management's interpretation, since management is paying for the new crew scheduling system
- **B.** Cancel the workshop until the union and management have resolved all of their disagreements
- **C.** Record both views, refer the dispute to the right forum and keep the rules configurable
- **D.** Choose the union's interpretation, since the union's members will use the system every day

Question 155. Which two types of information does a stakeholder engagement plan typically contain? (Choose two.)

- **A.** Desired engagement levels and strategies
- **B.** The log of defects found in testing
- **C.** Detailed cost estimates for activities
- **D.** The methods and timing of engagement
- **E.** The payment terms of the contracts

Question 156. The security team often blocks releases with findings at the last minute, and the developers are angry. What should the project manager do?

- **A.** Ask the security team to stop blocking releases, since delivery dates must always come first
- **B.** Ask the developers to accept the findings, since the security team has the final authority
- **C.** Bring both teams together to move security checks earlier and agree shared release criteria
- **D.** Escalate the conflict to the chief information officer and ask for a decision on who is right

Question 157. After go-live, a scheduler wants to adjust a crew rule parameter in the live system. What should happen?

- **A.** The scheduler makes the change directly, since schedulers use the system daily
- **B.** The vendor makes the change during its next scheduled maintenance window next month
- **C.** The change goes through the operational change process, with testing and approval
- **D.** The change is made and reported at the next monthly operations meeting

Question 158. Help desk staff solve the same problems again and again. Which practice captures solutions as part of the daily work?

- **A.** Writing a complete manual of all known solutions at the end of each year
- **B.** Creating and improving knowledge articles while resolving each ticket
- **C.** Recording the solutions in each staff member's personal notes
- **D.** Asking each new staff member to shadow a colleague for a month

Question 159. Before a plant restarts after modifications, which review confirms that the work matches the design and that the safety systems are in place?

- **A.** A sprint review
- **B.** A lessons learned review
- **C.** A pre-startup safety review
- **D.** A procurement audit

Question 160. How should acceptance of a data migration be defined?

- **A.** Through user complaints in the first month after go-live, since users notice errors
- **B.** Measurable reconciliation criteria agreed with the data owners
- **C.** Through the project manager's judgement of whether the data looks correct
- **D.** Through the vendor's confirmation that its migration scripts ran without any errors

Question 161. The regulator complains that the project's safety report was late and incomplete. What should the project manager do?

- **A.** Ask the regulator to accept the incomplete report, since most of the content is there
- **B.** Explain that the report was late because the team was busy with other important work
- **C.** Acknowledge the problem, provide the complete report quickly and agree a reporting schedule
- **D.** Send the report directly to the regulator's manager to avoid the complaining officer

Question 162. After each enrolment period, the team wants to learn quickly what worked and what did not. What should it hold?

- **A.** A survey of all students, analysed at the end of the year
- **B.** A lessons learned session at the end of the whole programme
- **C.** A short after-action review with the people involved
- **D.** A formal audit carried out by the internal audit department

Question 163. Which metrics best show whether the security operations centre is improving its incident response?

- **A.** Mean time to detect and mean time to respond
- **B.** Number of meetings held by the security team

- **C.** Number of analysts employed in the centre
- **D.** Number of security tools that have been bought

Question 164. A personal data breach has been confirmed. Which two actions does data protection law typically require? (Choose two.)

- **A.** Notifying the data protection authority within the legal deadline
- **B.** Asking the affected people to keep the breach secret
- **C.** Informing the affected people when the risk to them is high
- **D.** Deleting the logs that show how the breach happened
- **E.** Waiting to see whether the breach becomes public

Question 165. Leaders of an airline programme want a dedicated room where goals, progress and problems are visible to everyone involved. What is this practice called?

- **A.** A sandbox environment
- **B.** A war chest fund
- **C.** A data warehouse
- **D.** An obeya (big room)

Question 166. The crew app must work on many phone models, but the team has only a few test devices. What should the team do?

- **A.** Support only one phone model to keep testing simple
- **B.** Ask the crew members to test the app on their own phones after release
- **C.** Test only on the devices the team already owns
- **D.** Use a device lab or a cloud device-testing service

Question 167. Which combined log is often reviewed by steering committees to track risks, assumptions, issues and dependencies?

- **A.** A burndown chart
- **B.** A product backlog
- **C.** A RAID log
- **D.** A RACI chart

Question 168. An open-source tool meets 80% of the university's needs at a fraction of the cost of a custom system. What should the project consider?

- **A.** Rejecting the tool, since open-source software is always too risky for a university
- **B.** Waiting until the tool meets 100% of the needs before considering it any further
- **C.** Adopting the tool and adapting processes for the gaps, if the overall value is higher
- **D.** Building a custom system anyway, since only a custom system can meet 100% of the needs

Question 169. A conflict between two team members continues despite the scrum master's facilitation, and it is now affecting the whole team. What should the scrum master do?

- **A.** Ask the rest of the team to vote on who is right in the conflict
- **B.** Ignore the conflict, since the team should resolve it on its own
- **C.** Move one of the two team members to another team without discussion
- **D.** Involve a neutral mediator, such as an experienced coach or HR

Question 170. Before a major change to the student system, what should be prepared in case the change fails?

- **A.** A lessons learned report
- **B.** A tested backout plan
- **C.** A new project charter
- **D.** A press release

Question 171. A stakeholder is frustrated that the release forecast changes every sprint. What should the product owner do?

- **A.** Explain that forecasts are updated with new data, and show the range and its trend
- **B.** Give the stakeholder a single fixed date and ask the team to meet it whatever happens
- **C.** Stop updating the forecast, so that the stakeholder sees the same date every sprint
- **D.** Stop sharing the forecast with the stakeholder until the release date is certain

Question 172. After an accident at another chemical plant, local residents demand more information about Halden's safety. What should the project do?

- **A.** Avoid public statements until the media interest in the other accident has died down
- **B.** Engage the community proactively about the upgrade and its safety measures
- **C.** Release detailed technical drawings of the plant so that residents can judge for themselves
- **D.** Refer all questions to the regulator, since the regulator is responsible for public safety

Question 173. To define the scope of a new terminal building, the project manager breaks the product down into its components and analyses each one. Which technique is this?

- **A.** Benchmarking
- **B.** Product analysis
- **C.** Rolling wave planning
- **D.** Resource levelling

Question 174. Executives struggle with the new collaboration tools, while junior staff are experts in them. Which practice could help?

- **A.** Reverse mentoring
- **B.** A procurement audit
- **C.** A change freeze
- **D.** A performance review

Question 175. How detailed should the schedule be for a 21-day plant shutdown?

- **A.** Monthly milestones, since the shutdown is shorter than a month
- **B.** No schedule, since the work depends on what is found inside the equipment
- **C.** Very detailed, often down to hours, with shift-level coordination
- **D.** Weekly activities only, since daily detail changes too often

Question 176. At the end of a funded research project, the funder requires the data to be preserved and shared. What should the project do?

- **A.** Send the data by email to anyone who requests it, without any conditions or checks
- **B.** Deposit the data in an approved repository, as the data management plan requires

- **C.** Delete the data, since the research questions have now been answered
- **D.** Keep the data on the lead researcher's laptop in case anyone asks for it

Question 177. As the shutdown ends, contractor crews finish their work at different times. What should the project manager do?

- **A.** Demobilise crews as their work is accepted, checking tools, permits and badges
- **B.** Ask the crews to stay for an extra week in case the plant has problems after restart
- **C.** Let each crew leave whenever it wants, without checking its work or its equipment
- **D.** Keep all of the crews on site until the last crew has finished, so that they leave together

Question 178. A member of the evaluation panel discloses that her spouse works for one of the bidders. What should the project manager do?

- **A.** Disqualify the bidder, since a conflict of interest exists
- **B.** Remove her from the evaluation and record the disclosure
- **C.** Keep her on the panel, since she has been honest about it
- **D.** Ask her to give that bidder an average score to stay neutral

Question 179. The operating system used by the plant's control servers will lose vendor support next year. What should the project manager do?

- **A.** Keep the current system, since it works well and has never failed in fifteen years
- **B.** Wait until the support has ended before deciding what to do
- **C.** Disconnect the servers from the network permanently to avoid risk
- **D.** Plan an upgrade or replacement, and assess the risks until it is complete

Question 180. A process has a process capability index (Cpk) of 0.8. What does this indicate?

- **A.** The process is in statistical control and fully capable of meeting the specification
- **B.** The process cannot consistently meet the specification limits
- **C.** The specification limits are too wide
- **D.** The process produces no defects at all

Answer Key

Q	Answer	Domain	Task	Approach	Q	Answer	Domain	Task	Approach
1	**B**	People	PE1	Predictive	91	**C**	Bus. Env.	BE6	Agile
2	**A**	Process	PR1	Predictive	92	**B**	Process	PR7	Agile
3	**A**	Process	PR1	Hybrid	93	**C**	Process	PR6	Agile
4	**D**	Process	PR4	Agile	94	**C**	Bus. Env.	BE7	Agile
5	**B**	Bus. Env.	BE2	Hybrid	95	**B**	People	PE4	Agile
6	**A**	Process	PR10	Hybrid	96	**C**	Bus. Env.	BE5	Agile
7	**C**	Process	PR7	Agile	97	**3**	Process	PR8	Predictive
8	**A**	People	PE6	Hybrid	98	**D**	Bus. Env.	BE6	Agile
9	**C**	People	PE7	Agile	99	**A**	People	PE5	Hybrid
10	**D**	People	PE6	Predictive	100	**A**	Process	PR9	Agile
11	**D**	Bus. Env.	BE3	Agile	101	**A**	People	PE5	Hybrid
12	**A**	People	PE8	Predictive	102	**B**	Process	PR1	Predictive
13	**D**	People	PE7	Predictive	103	**A**	Process	PR8	Predictive
14	**B**	Process	PR3	Predictive	104	**B**	Bus. Env.	BE7	Agile
15	**A**	Process	PR9	Hybrid	105	**B**	People	PE1	Hybrid
16	**B**	People	PE2	Predictive	106	**A**	Process	PR5	Hybrid
17	**B**	People	PE5	Predictive	107	**B, E**	Process	PR8	Agile
18	**B**	Process	PR3	Agile	108	**1→B, 2→C, 3→D, 4→A**	Process	PR5	Predictive
19	**B**	People	PE3	Agile	109	**1→A, 2→B, 3→D, 4→C**	People	PE3	Predictive
20	**C**	Process	PR5	Agile	110	**B**	People	PE5	Predictive
21	**A**	Process	PR10	Predictive	111	**A**	Process	PR2	Agile
22	**B**	Process	PR5	Hybrid	112	**C**	Bus. Env.	BE4	Predictive
23	**1→A, 2→C, 3→D, 4→B**	Bus. Env.	BE7	Predictive	113	**C, E**	Process	PR1	Agile

Q	Answer	Domain	Task	Approach	Q	Answer	Domain	Task	Approach
24	**C**	Process	PR9	Predictive	114	**B**	Bus. Env.	BE3	Predictive
25	**B**	People	PE8	Predictive	115	**D**	Bus. Env.	BE8	Agile
26	**C**	Bus. Env.	BE2	Hybrid	116	**A**	People	PE1	Agile
27	**B**	People	PE8	Hybrid	117	**C**	Process	PR6	Predictive
28	**B**	Process	PR5	Predictive	118	**C**	People	PE1	Agile
29	**C**	People	PE6	Predictive	119	**D**	People	PE8	Agile
30	**C**	Process	PR7	Predictive	120	**D**	Process	PR2	Agile
31	**C**	Process	PR4	Hybrid	121	**B**	People	PE8	Agile
32	**D**	Process	PR9	Predictive	122	**C**	People	PE4	Hybrid
33	**A**	Process	PR1	Hybrid	123	**D**	Bus. Env.	BE1	Predictive
34	**A**	People	PE4	Agile	124	**B**	Bus. Env.	BE3	Predictive
35	**D**	Process	PR2	Agile	125	**B**	Bus. Env.	BE5	Hybrid
36	**A**	Bus. Env.	BE4	Hybrid	126	**C**	Bus. Env.	BE7	Hybrid
37	**C**	Bus. Env.	BE2	Predictive	127	**C**	Process	PR1	Agile
38	**A**	Process	PR7	Hybrid	128	**12**	People	PE8	Predictive
39	**C**	People	PE8	Hybrid	129	**D**	Process	PR9	Hybrid
40	**A**	Bus. Env.	BE1	Agile	130	**D**	People	PE4	Predictive
41	**D**	Bus. Env.	BE1	Hybrid	131	**D**	Process	PR9	Predictive
42	**D**	Bus. Env.	BE8	Hybrid	132	**D**	Process	PR10	Predictive
43	**C**	Bus. Env.	BE2	Predictive	133	**D**	Process	PR3	Hybrid
44	**C**	People	PE3	Predictive	134	**A, D**	People	PE1	Hybrid
45	**D**	Process	PR2	Predictive	135	**B**	Bus. Env.	BE6	Predictive
46	**A**	People	PE5	Agile	136	**D**	People	PE4	Hybrid
47	**C**	Process	PR1	Agile	137	**A**	People	PE3	Agile

Q	Answer	Domain	Task	Approach	Q	Answer	Domain	Task	Approach
48	**A, B**	Bus. Env.	BE4	Predictive	138	**A**	Bus. Env.	BE7	Hybrid
49	**A, B**	People	PE3	Hybrid	139	**A**	Process	PR3	Predictive
50	**B**	People	PE3	Hybrid	140	**B**	Process	PR4	Hybrid
51	**D**	Process	PR2	Hybrid	141	**C**	Bus. Env.	BE4	Agile
52	**D**	Bus. Env.	BE7	Predictive	142	**A**	Process	PR6	Hybrid
53	**D**	Bus. Env.	BE4	Hybrid	143	**B**	Process	PR5	Hybrid
54	**B**	Process	PR4	Predictive	144	**B**	People	PE5	Agile
55	**$450,000**	Process	PR6	Predictive	145	**A**	Process	PR2	Predictive
56	**B**	People	PE6	Agile	146	**D**	People	PE3	Agile
57	**C**	People	PE3	Predictive	147	**A**	Process	PR6	Hybrid
58	**A**	Process	PR6	Predictive	148	**C**	Process	PR8	Predictive
59	**B**	People	PE2	Hybrid	149	**A**	Bus. Env.	BE5	Predictive
60	**C**	Bus. Env.	BE1	Predictive	150	**C**	Process	PR3	Agile
61	**D**	Process	PR4	Predictive	151	**D**	People	PE1	Predictive
62	**C**	People	PE6	Hybrid	152	**D**	Bus. Env.	BE4	Agile
63	**C**	People	PE2	Agile	153	**D**	Bus. Env.	BE5	Predictive
64	**A**	Process	PR4	Agile	154	**C**	People	PE2	Hybrid
65	**D**	People	PE2	Predictive	155	**A, D**	People	PE4	Predictive
66	**A**	People	PE4	Predictive	156	**C**	People	PE2	Agile
67	**B**	Process	PR8	Agile	157	**C**	Bus. Env.	BE3	Predictive
68	**D**	Process	PR5	Predictive	158	**B**	People	PE7	Agile
69	**D**	Bus. Env.	BE1	Hybrid	159	**C**	Process	PR7	Predictive
70	**A**	Bus. Env.	BE6	Predictive	160	**B**	Process	PR2	Hybrid
71	**A**	Bus. Env.	BE3	Agile	161	**C**	People	PE6	Predictive

Q	Answer	Domain	Task	Approach	Q	Answer	Domain	Task	Approach
72	**B**	Bus. Env.	BE8	Predictive	162	**C**	People	PE7	Hybrid
73	**D**	Process	PR1	Predictive	163	**A**	Process	PR9	Hybrid
74	**D**	People	PE3	Hybrid	164	**A, C**	Bus. Env.	BE2	Hybrid
75	**C**	People	PE5	Predictive	165	**D**	People	PE1	Hybrid
76	**A**	Process	PR6	Agile	166	**D**	Process	PR4	Hybrid
77	**D**	Bus. Env.	BE6	Hybrid	167	**C**	Bus. Env.	BE1	Predictive
78	**A, C**	Process	PR3	Predictive	168	**C**	Process	PR3	Hybrid
79	**A**	Process	PR6	Predictive	169	**D**	People	PE2	Agile
80	**D**	Bus. Env.	BE2	Agile	170	**B**	Bus. Env.	BE3	Hybrid
81	**B**	Process	PR7	Hybrid	171	**A**	People	PE6	Agile
82	**D**	Process	PR8	Hybrid	172	**B**	Bus. Env.	BE8	Hybrid
83	**C**	Bus. Env.	BE5	Agile	173	**B**	Process	PR2	Predictive
84	**A**	People	PE4	Predictive	174	**A**	People	PE7	Hybrid
85	**A**	Process	PR8	Agile	175	**C**	Process	PR8	Predictive
86	**D**	People	PE7	Predictive	176	**B**	Process	PR10	Hybrid
87	**C**	Bus. Env.	BE5	Hybrid	177	**A**	Process	PR10	Predictive
88	**A**	Process	PR10	Agile	178	**B**	People	PE2	Predictive
89	**$85,000**	Bus. Env.	BE5	Predictive	179	**D**	Bus. Env.	BE8	Predictive
90	**B**	Process	PR3	Agile	180	**B**	Process	PR7	Predictive

Detailed Explanations

Q1 — Answer: B · *People · PE1 Develop a common vision · Predictive*

Correct: B. A project canvas

A project canvas summarises the essential elements of a project on one page and is created collaboratively, which builds shared understanding.

Q2 — Answer: A · *Process · PR1 Develop an integrated project management plan and plan delivery · Predictive*

Correct: A. An inspection and test plan

An inspection and test plan lists the inspections, tests, acceptance criteria and hold or witness points for each stage of the work.

Q3 — Answer: A · *Process · PR1 Develop an integrated project management plan and plan delivery · Hybrid*

Correct: A. Integrate change management activities into the plan

Delivering value requires that people adopt the solution. Training, communication and support are planned and resourced as part of the integrated plan.

Q4 — Answer: D · *Process · PR4 Plan and manage resources · Agile*

Correct: D. Let the team decide how to split, keeping the skills balanced across both teams

Involving the team in the split builds ownership, and balancing skills keeps both teams cross-functional.

Q5 — Answer: B · *Business Environment · BE2 Plan and manage project compliance · Hybrid*

Correct: B. Decline and consult legal, since sharing such information may break competition law

Sharing competitively sensitive information, such as future prices, with competitors can breach competition (antitrust) law.

Q6 — Answer: A · *Process · PR10 Manage project closure · Hybrid*

Correct: A. Transferring each control to a named operational owner, with procedures and metrics

Sustaining value after closure requires clear operational ownership, documented procedures and measures of performance.

Q7 — Answer: C · *Process · PR7 Plan and optimize quality of products/deliverables · Agile*

Correct: C. About five

Small rounds of usability testing, with about five users, find most problems. Several small rounds are better than one large one.

Q8 — Answer: A · *People · PE6 Manage stakeholder expectations · Hybrid*

Correct: A. Acknowledge the concern, explain why, and reduce the friction where possible

Stakeholders' concerns are taken seriously. Explaining the reasons and improving the experience, for example with remembered devices, maintains both security and support.

Q9 — Answer: C · *People · PE7 Help ensure knowledge transfer · Agile*

Correct: C. Write and maintain response playbooks, and rotate analysts through incidents in pairs

Playbooks capture explicit knowledge, while pairing on real incidents transfers tacit knowledge and removes single points of failure.

Q10 — Answer: D · *People · PE6 Manage stakeholder expectations · Predictive*

Correct: D. Find the root cause of the recurring defects, fix the process and share the improvement plan

Recurring defects point to a process problem. Fixing the cause and sharing the plan restores the customer's confidence.

Q11 — Answer: D · *Business Environment · BE3 Manage and control changes · Agile*

Correct: D. The product owner adds it to the product backlog and orders it so that it is done in time

New requirements enter the product backlog, where the product owner orders them by value and urgency, usually for an upcoming sprint.

Q12 — Answer: A · *People · PE8 Plan and manage communication · Predictive*

Correct: A. Confirm the agreement in writing through the agreed channel

Important agreements, especially contractual ones, are confirmed in writing through the agreed channels to avoid later disputes.

Q13 — Answer: D · *People · PE7 Help ensure knowledge transfer · Predictive*

Correct: D. Hypercare

Hypercare is a period of intensive post-go-live support, during which knowledge is transferred to the people who will support the system.

Q14 — Answer: B · *Process · PR3 Help ensure value-based delivery · Predictive*

Correct: B. As reduced risk and avoided losses, plus compliance and reputational benefits

Safety investments create value by reducing the likelihood and impact of incidents, meeting obligations and protecting reputation.

Q15 — Answer: A · *Process · PR9 Evaluate project status · Hybrid*

Correct: A. Inspect the actual test results and deliverables to verify the status

Status should be verified against evidence. Inspecting actual results and deliverables reveals the real situation.

Q16 — Answer: B · *People · PE2 Manage conflicts · Predictive*

Correct: B. Compromising

Compromising, in which each side gives up something, is appropriate when parties have equal power and a quick, temporary solution is needed.

Q17 — Answer: B · *People · PE5 Align stakeholder expectations · Predictive*

Correct: B. Clarify that client-requested changes go through the contract's change process

A fixed price covers the agreed scope. Changes requested by the client are handled through the contract's change process, with their cost and time impacts.

Q18 — Answer: B · *Process · PR3 Help ensure value-based delivery · Agile*

Correct: B. A fake door test

A fake door test measures real interest before building a feature. Users who click are told that the feature is coming, so trust is maintained.

Q19 — Answer: B · *People · PE3 Lead the project team · Agile*

Correct: B. A short, timeboxed hackathon on ideas chosen by the team

Giving the team autonomy to explore its own ideas in a short, focused event can restore motivation and generate useful innovations.

Q20 — Answer: C · *Process · PR5 Plan and manage procurement · Agile*

Correct: C. Exit and data portability provisions

Exit and portability provisions reduce lock-in by making sure that data can be retrieved and services transferred.

Q21 — Answer: A · *Process · PR10 Manage project closure · Predictive*

Correct: A. That the acceptance criteria have been met

Formal acceptance against the agreed criteria comes before final payment and contract closure.

Q22 — Answer: B · *Process · PR5 Plan and manage procurement · Hybrid*

Correct: B. A framework agreement with pre-selected suppliers

Framework agreements set terms with selected suppliers in advance, so that repeat purchases can be made quickly within the rules.

Q23 — Answer: 1→A, 2→C, 3→D, 4→B · *Business Environment · BE7 Support organizational change · Predictive*

Correct pairs: 1. ADKAR → The stages an individual goes through to change; 2. Kotter's eight steps → The steps leaders take to drive change in an organisation; 3. Lewin's three stages → Unfreezing, changing and refreezing behaviour; 4. Bridges' transition model → Endings, the neutral zone and new beginnings

Different change models highlight different aspects of change: individual stages, leadership steps, organisational stages or people's psychological transitions.

Q24 — Answer: C · *Process · PR9 Evaluate project status · Predictive*

Correct: C. Cost performance is poor, with a CPI of about 0.71, so an overrun is likely

CPI = EV / AC = 50% of BAC / 70% of BAC ≈ 0.71. Less than a dollar of value is earned for each dollar spent.

Q25 — Answer: B · *People · PE8 Plan and manage communication · Predictive*

Correct: B. Message mapping

Message mapping prepares clear, consistent key messages and supporting facts for sensitive or high-stakes communication.

Q26 — Answer: C · *Business Environment · BE2 Plan and manage project compliance · Hybrid*

Correct: C. Agree a copyright and licence check with the library before publishing

Copyright compliance is built into the process, with the right experts, so that lecturers can share material lawfully.

Q27 — Answer: B · *People · PE8 Plan and manage communication · Hybrid*

Correct: B. On a need-to-know basis, coordinated with legal and communications

Incident communication is controlled and coordinated so that the investigation is protected and legal obligations, such as notifications, are met.

Q28 — Answer: B · *Process · PR5 Plan and manage procurement · Predictive*

Correct: B. Terminate for cause, following the contract's notice procedure

Serious or repeated breaches can justify termination for cause (default), but the contract's procedures, such as notices and cure periods, must be followed.

Q29 — Answer: C · *People · PE6 Manage stakeholder expectations · Predictive*

Correct: C. Communicate quickly and honestly, agree fair remedies and investigate the cause

When a delivered solution fails stakeholders, prompt and honest communication, fair remedies and a root cause investigation help restore trust.

Q30 — Answer: C · *Process · PR7 Plan and optimize quality of products/deliverables · Predictive*

Correct: C. Analyse the error types, fix the data at source with its owners and repeat the trial

Data quality problems are analysed and fixed at the source, and trial migrations are repeated until the agreed quality criteria are met. Errors in crew qualifications could create safety and legal problems.

Q31 — Answer: C · *Process · PR4 Plan and manage resources · Hybrid*

Correct: C. Agree with operations a rota and backfill that release schedulers at planned times

Scarce business experts are planned as resources: their time is agreed with their managers, and backfill is arranged so that operations are not put at risk.

Q32 — Answer: D · *Process · PR9 Evaluate project status · Predictive*

Correct: D. A TCPI of 1.60, which is unrealistic, so a revised estimate at completion is needed

TCPI = (BAC − EV) / (BAC − AC) = 800,000 / 500,000 = 1.60. With a current CPI of 0.80, a 60% improvement in efficiency is not realistic, so the forecast should be revised.

Q33 — Answer: A · *Process · PR1 Develop an integrated project management plan and plan delivery · Hybrid*

Correct: A. Pre-agreed go/no-go criteria, such as roster accuracy and rule compliance

Go/no-go decisions are based on criteria agreed in advance with the stakeholders, which makes the decision objective and transparent.

Q34 — Answer: A · *People · PE4 Engage stakeholders · Agile*

Correct: A. Appoint security champions in each unit and involve them in prioritising the work

Champions inside each unit bring local knowledge and influence, and involving them in prioritisation builds ownership.

Q35 — Answer: D · *Process · PR2 Develop and manage project scope · Agile*

Correct: D. By source system, delivering one import at a time

Splitting by data source creates smaller, independently valuable stories that can be delivered and tested one at a time.

Q36 — Answer: A · *Business Environment · BE4 Remove impediments and manage issues · Hybrid*

Correct: A. Reopen it, analyse the root cause and verify the fix before closing it again

A recurring issue shows that the earlier fixes treated symptoms. Root cause analysis and verification prevent further recurrence.

Q37 — Answer: C · *Business Environment · BE2 Plan and manage project compliance · Predictive*

Correct: C. Implement them as requirements and test them against the regulations before go-live

Regulatory rules become requirements, verified through testing against the regulations themselves before the system is used.

Q38 — Answer: A · *Process · PR7 Plan and optimize quality of products/deliverables · Hybrid*

Correct: A. Keep automated regression tests for the integrations and run them on each release

Automated regression tests detect problems caused by external changes quickly, before they affect operations.

Q39 — Answer: C · *People · PE8 Plan and manage communication · Hybrid*

Correct: C. A cutover command centre with a communication schedule and clear escalation

A cutover command centre provides real-time coordination, regular checkpoints and fast escalation during a high-risk event.

Q40 — Answer: A · *Business Environment · BE1 Define and establish project governance · Agile*

Correct: A. A forum that agrees priorities against shared objectives, with a final say

Shared capacity is allocated through transparent governance that compares requests against common objectives, with clear authority for the final decision.

Q41 — Answer: D · *Business Environment · BE1 Define and establish project governance · Hybrid*

Correct: D. The risk owner with the authority to accept it, as governance defines

Accepting residual risk is a governance decision made by someone with the authority and accountability to do so.

Q42 — Answer: D · *Business Environment · BE8 Evaluate external business environment changes · Hybrid*

Correct: D. Reassess the business case and adjust scope or phasing with the sponsor

External changes that affect demand change the business case. The sponsor decides on adjustments based on an updated assessment.

Q43 — Answer: C · *Business Environment · BE2 Plan and manage project compliance · Predictive*

Correct: C. A permit to work, with isolation and gas testing confirmed

A permit-to-work system controls hazardous work by confirming isolations, testing and precautions before work starts.

Q44 — Answer: C · *People · PE3 Lead the project team · Predictive*

Correct: C. Self-awareness

Self-awareness is recognising one's own emotions and their effect on behaviour. It is the foundation for self-regulation.

Q45 — Answer: D · *Process · PR2 Develop and manage project scope · Predictive*

Correct: D. By major deliverables, by phases or by subprojects, depending on the project

The WBS can be structured in different ways, such as by deliverables, phases or subprojects, as long as it covers 100% of the scope.

Q46 — Answer: A · *People · PE5 Align stakeholder expectations · Agile*

Correct: A. Communicate the freeze and its reasons in advance, and show when the features will come

Explaining constraints in advance, with a clear view of when features will arrive, keeps expectations realistic.

Q47 — Answer: C · *Process · PR1 Develop an integrated project management plan and plan delivery · Agile*

Correct: C. Every sprint, based on the latest velocity and backlog changes

Release plans are living plans. They are revised every sprint as the team learns more about its pace and the backlog changes.

Q48 — Answer: A, B · *Business Environment · BE4 Remove impediments and manage issues · Predictive*

Correct: A. The urgency of resolving the issue / B. The impact on the project's objectives

Issues are prioritised by their impact and urgency, so that the most important problems are resolved first.

Q49 — Answer: A, B · *People · PE3 Lead the project team · Hybrid*

Correct: A. Sharing what is known and unknown, honestly and often / B. Focusing the team on what it can control

Honest, frequent communication and a focus on what the team can influence maintain trust and productivity when much is uncertain.

Q50 — Answer: B · *People · PE3 Lead the project team · Hybrid*

Correct: B. Discuss their career goals and offer development opportunities and recognition

Retention depends on growth, recognition and meaningful work. Understanding people's career goals lets the leader address what matters to them.

Q51 — Answer: D · *Process · PR2 Develop and manage project scope · Hybrid*

Correct: D. Use cases with main and alternative flows

Use cases describe the interaction step by step, including alternative and exception flows, which suits complex processes.

Q52 — Answer: D · *Business Environment · BE7 Support organizational change · Predictive*

Correct: D. Schedule training for every shift, including nights and weekends

Change activities must reach everyone affected, which means fitting training to how people actually work.

Q53 — Answer: D · *Business Environment · BE4 Remove impediments and manage issues · Hybrid*

Correct: D. Agree delegates with authority through the faculty's governance

Decision bottlenecks are resolved by agreeing properly authorised delegates, not by bypassing approvals.

Q54 — Answer: B · *Process · PR4 Plan and manage resources · Predictive*

Correct: B. Mobilisation logistics, such as inductions, accommodation, permits and supervision

Large temporary workforces need detailed planning of mobilisation, induction, logistics and supervision, as well as numbers.

Q55 — Answer: $450,000 · *Process · PR6 Plan and manage finance · Predictive*

ETC = EAC − AC = 750,000 − 300,000 = $450,000.

Q56 — Answer: B · *People · PE6 Manage stakeholder expectations · Agile*

Correct: B. Agree response times for decisions and secure the product owner's availability

Vendors are stakeholders too. Slow decisions are an impediment that the project manager addresses by agreeing expectations and securing availability.

Q57 — Answer: C · *People · PE3 Lead the project team · Predictive*

Correct: C. Promote a just culture: thank reporters, focus on systems and act on reports

A just culture separates honest mistakes from reckless behaviour, makes reporting safe and shows that reports lead to improvements.

Q58 — Answer: A · *Process · PR6 Plan and manage finance · Predictive*

Correct: A. The 50/50 rule

The 50/50 rule is a fixed-formula method for short activities. It is simple, but it can overstate progress on activities that have started but are far from finished.

Q59 — Answer: B · *People · PE2 Manage conflicts · Hybrid*

Correct: B. Facilitate a discussion to find solutions meeting both needs, such as approved exceptions

Collaborative problem solving looks for options that satisfy both interests, such as controlled exceptions with compensating safeguards.

Q60 — Answer: C · *Business Environment · BE1 Define and establish project governance · Predictive*

Correct: C. The operational owner, against agreed operational acceptance criteria

Operational acceptance is given by the business owner who will run the system, based on agreed criteria such as support readiness and performance.

Q61 — Answer: D · *Process · PR4 Plan and manage resources · Predictive*

Correct: D. That the certifications are verified before the workers are mobilised

Competence for safety-critical work is verified before people start, as part of acquiring resources.

Q62 — Answer: C · *People · PE6 Manage stakeholder expectations · Hybrid*

Correct: C. Offer that member a separate technical briefing, and keep board updates at the right level

Different stakeholders need different levels of detail. A separate briefing meets the member's needs without overloading the board.

Q63 — Answer: C · *People · PE2 Manage conflicts · Agile*

Correct: C. Help the team discuss its needs and agree a working agreement

Working agreements created by the team resolve recurring disagreements about how the team works, and people commit to rules they help create.

Q64 — Answer: A · *Process · PR4 Plan and manage resources · Agile*

Correct: A. Typically 10 or fewer people

The Scrum Guide describes Scrum teams as typically 10 or fewer people: small enough to stay nimble and large enough to complete significant work.

Q65 — Answer: D · *People · PE2 Manage conflicts · Predictive*

Correct: D. Anchoring

Anchoring uses a first offer to set a reference point. Knowing the market and one's own BATNA helps a negotiator resist it.

Q66 — Answer: A · *People · PE4 Engage stakeholders · Predictive*

Correct: A. Local emergency services and the authorities responsible for emergency planning

Projects at major hazard sites affect emergency planning outside the site, so emergency services and the responsible authorities are key stakeholders.

Q67 — Answer: B · *Process · PR8 Plan and manage schedule · Agile*

Correct: B. Yes; increments can be released at any time, and the sprint is a planning cadence

Scrum does not restrict releases to the end of a sprint. The sprint provides the rhythm for planning, review and improvement.

Q68 — Answer: D · *Process · PR5 Plan and manage procurement · Predictive*

Correct: D. Their safety performance and qualifications, through pre-qualification

Pre-qualification makes sure that only competent, safe contractors are invited to bid for high-risk work.

Q69 — Answer: D · *Business Environment · BE1 Define and establish project governance · Hybrid*

Correct: D. A service owner, supported by a board representing the faculties

Shared services need clear ownership and representative governance, so that priorities are set fairly and transparently.

Q70 — Answer: A · *Business Environment · BE6 Continuous improvement · Predictive*

Correct: A. Review the published findings and apply the relevant lessons to the design and procedures

Learning from external incidents is a powerful improvement practice, especially in high-hazard industries.

Q71 — Answer: A · *Business Environment · BE3 Manage and control changes · Agile*

Correct: A. The product owner reprioritises the backlog, using the agreed expedite route

Urgent, high-value work is handled through the backlog and the team's agreed policies for expedited work, so that priorities stay transparent.

Q72 — Answer: B · *Business Environment · BE8 Evaluate external business environment changes · Predictive*

Correct: B. Assess the impact on operations and the project with logistics and compliance

Regulatory changes in the wider business environment can affect a project's design and value, so they are assessed early with the relevant experts.

Q73 — Answer: D · *Process · PR1 Develop an integrated project management plan and plan delivery · Predictive*

Correct: D. A work authorisation system

A work authorisation system controls when and by whom work packages are started, which keeps execution aligned with the plan.

Q74 — Answer: D · *People · PE3 Lead the project team · Hybrid*

Correct: D. Establish clear command: one incident lead, defined roles and a regular update rhythm

In a crisis, a directive style with clear command, roles and communication rhythm reduces confusion and speeds up the response.

Q75 — Answer: C · *People · PE5 Align stakeholder expectations · Predictive*

Correct: C. Explain the approach and how archived data can be accessed, and agree it with the owner

Expectation gaps are closed by explaining the approach and its reasons, showing how needs will still be met and getting the owner's agreement.

Q76 — Answer: A · *Process · PR6 Plan and manage finance · Agile*

Correct: A. The release is likely to overrun its budget unless the scope or the plan changes

Comparing spending with delivered value shows the cost trend. Early action, such as reprioritising scope, can keep the release within budget.

Q77 — Answer: D · *Business Environment · BE6 Continuous improvement · Hybrid*

Correct: D. Gather feedback, simplify the templates and measure the effect

Continuous improvement applies to the organisation's own processes: feedback, small changes and measurement improve them.

Q78 — Answer: A, C · *Process · PR3 Help ensure value-based delivery · Predictive*

Correct: A. A stronger brand reputation / C. Improved staff morale

Intangible benefits are hard to measure in money, such as morale or reputation. Cost reductions and revenue are tangible benefits.

Q79 — Answer: A · *Process · PR6 Plan and manage finance · Predictive*

Correct: A. Activities, work packages, control accounts, project

Activity costs roll up into work packages, then into control accounts, and finally into the project's cost baseline, together with contingency reserves.

Q80 — Answer: D · *Business Environment · BE2 Plan and manage project compliance · Agile*

Correct: D. Switch to an approved tool, or request approval through the proper process

Unapproved tools can breach security and data rules. Teams use approved tools or obtain approval, even when it is less convenient.

Q81 — Answer: B · *Process · PR7 Plan and optimize quality of products/deliverables · Hybrid*

Correct: B. Penetration testing

Penetration testing simulates real attacks to find exploitable weaknesses before attackers do.

Q82 — Answer: D · *Process · PR8 Plan and manage schedule · Hybrid*

Correct: D. Baseline the predictive work, forecast the agile work from velocity and link them through milestones

Hybrid projects tailor control: baselines where scope is stable, empirical forecasts where it evolves, and shared milestones to connect them.

Q83 — Answer: C · *Business Environment · BE5 Plan and manage risk · Agile*

Correct: C. On the areas with the highest risk, based on the likelihood and impact of failure

Risk-based testing directs limited effort to where failures are most likely and most harmful.

Q84 — Answer: A · *People · PE4 Engage stakeholders · Predictive*

Correct: A. Meet her one-to-one to understand her concerns

Understanding the reasons for resistance, in a private conversation, is the first step towards resolving it.

Q85 — Answer: A · *Process · PR8 Plan and manage schedule · Agile*

Correct: A. An expedite class of service with its own policy

Classes of service define how different types of work are treated. An expedite class allows urgent items to move ahead under explicit rules.

Q86 — Answer: D · *People · PE7 Help ensure knowledge transfer · Predictive*

Correct: D. Regular drills and tabletop exercises

Drills and tabletop exercises let people practise rare procedures, which keeps the knowledge alive and reveals gaps in the plans.

Q87 — Answer: C · *Business Environment · BE5 Plan and manage risk · Hybrid*

Correct: C. Regular backups with tested recovery procedures

Backups reduce the impact of data loss, but only tested recovery procedures prove that data can actually be restored.

Q88 — Answer: A · *Process · PR10 Manage project closure · Agile*

Correct: A. They are not done, and they go back to the product backlog for the product owner

Only items that meet the definition of done count as complete. Unfinished items go back to the product backlog.

Q89 — Answer: $85,000 · *Business Environment · BE5 Plan and manage risk · Predictive*

EMV = 0.10 × 200,000 + 0.30 × 50,000 + 0.05 × 1,000,000 = 20,000 + 15,000 + 50,000 = $85,000.

Q90 — Answer: B · *Process · PR3 Help ensure value-based delivery · Agile*

Correct: B. By the risk reduction each control achieves relative to its cost

Value-based prioritisation in security compares the risk reduction with the cost, so that the budget buys the most protection.

Q91 — Answer: C · *Business Environment · BE6 Continuous improvement · Agile*

Correct: C. Chaos engineering

Chaos engineering, sometimes run as "game days", tests resilience through controlled experiments and leads to improvements.

Q92 — Answer: B · *Process · PR7 Plan and optimize quality of products/deliverables · Agile*

Correct: B. Focus on meaningful tests and use coverage as an indicator, not a target

When a measure becomes a target, people optimise the number rather than quality. Coverage is useful as one indicator among others.

Q93 — Answer: C · *Process · PR6 Plan and manage finance · Agile*

Correct: C. $60,000

Cost of delay = value per week × weeks of delay = 20,000 × 3 = $60,000.

Q94 — Answer: C · *Business Environment · BE7 Support organizational change · Agile*

Correct: C. Regular, realistic simulations with immediate, supportive training

Behaviour changes through frequent practice and immediate, constructive feedback, not through shame or one-off training.

Q95 — Answer: B · *People · PE4 Engage stakeholders · Agile*

Correct: B. Use short mobile surveys, in-app feedback and sessions at crew bases

Engagement methods should fit how stakeholders work. Short, mobile and on-site options reach users with irregular schedules.

Q96 — Answer: C · *Business Environment · BE5 Plan and manage risk · Agile*

Correct: C. Assess the risk and plan responses, such as preparing an alternative

Dependencies on fragile external components are risks. Responses can include monitoring, contributing to the library or preparing alternatives.

Q97 — Answer: 3 · *Process · PR8 Plan and manage schedule · Predictive*

PERT standard deviation = (P − O) / 6 = (22 − 4) / 6 = 3 days.

Q98 — Answer: D · *Business Environment · BE6 Continuous improvement · Agile*

Correct: D. The improvement kata

The improvement kata sets a target condition and approaches it through repeated small experiments and reflection.

Q99 — Answer: A · *People · PE5 Align stakeholder expectations · Hybrid*

Correct: A. Explain why the shutdown is needed and plan it with operations, e.g. during scheduled maintenance

Constraints that conflict with expectations are explained early, and solutions are found together, such as aligning with planned maintenance.

Q100 — Answer: A · *Process · PR9 Evaluate project status · Agile*

Correct: A. Stories are finished only at the end; the team should split work and finish items steadily

A late drop suggests mini-waterfalls within the sprint. Smaller stories and finishing items one at a time improve flow and reduce risk.

Q101 — Answer: A · *People · PE5 Align stakeholder expectations · Hybrid*

Correct: A. Explain that risk can be reduced but not eliminated, and agree measurable targets

Expectations must be realistic. Agreeing measurable targets for risk reduction gives the board a meaningful way to judge success.

Q102 — Answer: B · *Process · PR1 Develop an integrated project management plan and plan delivery · Predictive*

Correct: B. Months in advance, with an early scope freeze and a detailed shutdown schedule

Shutdowns are short, expensive and complex. They need long preparation, an early scope freeze and very detailed schedules.

Q103 — Answer: A · *Process · PR8 Plan and manage schedule · Predictive*

Correct: A. Start-to-finish

In a start-to-finish relationship, the successor cannot finish until the predecessor has started. It is rare, but shift handovers are a classic example.

Q104 — Answer: B · *Business Environment · BE7 Support organizational change · Agile*

Correct: B. Funding persistent teams and defining product ownership roles

Moving to a product operating model requires changes to funding, roles and governance, not just new titles.

Q105 — Answer: B · *People · PE1 Develop a common vision · Hybrid*

Correct: B. Connect the programme to real incidents and to everyone going home safely

People commit to a goal when they understand why it matters to them. Linking the programme to real consequences builds purpose far better than rules or rewards.

Q106 — Answer: A · *Process · PR5 Plan and manage procurement · Hybrid*

Correct: A. Service levels, such as detection and response times, with reporting and remedies

Service contracts define measurable service levels, reporting and remedies, so that performance can be managed.

Q107 — Answer: B, E · *Process · PR8 Plan and manage schedule · Agile*

Correct: B. Monte Carlo simulation based on throughput / E. Velocity-based forecasting that gives a range

Forecasts based on actual performance, expressed as ranges or probabilities, are more reliable than wishes or best-case assumptions.

Q108 — Answer: 1→B, 2→C, 3→D, 4→A · *Process · PR5 Plan and manage procurement · Predictive*

Correct pairs: 1. Firm fixed price → The price is fixed for the defined scope, and the seller bears the cost risk; 2. Cost plus award fee → Costs are reimbursed, plus a fee based on the buyer's judgement of performance; 3. Time and materials → The buyer pays agreed rates for time and materials used; 4. Fixed price with economic price adjustment → The price is adjusted in line with defined indexes, such as material costs

The contract type allocates risk between buyer and seller and should fit how well the scope is defined.

Q109 — Answer: 1→A, 2→B, 3→D, 4→C · *People · PE3 Lead the project team · Predictive*

Correct pairs: 1. Maslow → A hierarchy of needs, from basic needs to self-actualisation; 2. Herzberg → Hygiene factors prevent dissatisfaction, while motivators drive satisfaction; 3. McClelland → Needs for achievement, affiliation and power; 4. Vroom → Motivation depends on expectancy, instrumentality and valence

Understanding motivation theories helps leaders choose suitable ways to engage different team members.

Q110 — Answer: B · *People · PE5 Align stakeholder expectations · Predictive*

Correct: B. Explain that the warranty covers defects; new features are changes

Expectations are aligned against the contract: warranties cover defects, while new functionality is new scope.

Q111 — Answer: A · *Process · PR2 Develop and manage project scope · Agile*

Correct: A. Regularly remove or archive items that are no longer relevant

A backlog is useful only if it is manageable. Regular pruning keeps it focused on what is still valuable.

Q112 — Answer: C · *Business Environment · BE4 Remove impediments and manage issues · Predictive*

Correct: C. Work with the supplier and customs agent to supply the documents, and assess the impact

Issues are resolved by working with the parties involved to remove the cause, while assessing and managing the impact on the project.

Q113 — Answer: C, E · *Process · PR1 Develop an integrated project management plan and plan delivery · Agile*

Correct: C. The ordered product backlog / E. The team's capacity and past performance

Sprint planning uses the ordered product backlog, the latest increment and the team's capacity and past performance to select work and agree a sprint goal.

Q114 — Answer: B · *Business Environment · BE3 Manage and control changes · Predictive*

Correct: B. Its impact on the project's objectives and value, compared with its costs and risks

Change decisions weigh the change's contribution to objectives and value against its impact on scope, schedule, cost, quality and risk.

Q115 — Answer: D · *Business Environment · BE8 Evaluate external business environment changes · Agile*

Correct: D. Assess the cost impact and options, such as optimisation or alternative providers

Market changes are analysed for their impact on cost and value, and options are compared before deciding.

Q116 — Answer: A · *People · PE1 Develop a common vision · Agile*

Correct: A. Make security part of the team's goals and definition of done, with experts in planning

Shared ownership grows when security is part of the team's own goals and standards, and when experts collaborate with the team from the start.

Q117 — Answer: C · *Process · PR6 Plan and manage finance · Predictive*

Correct: C. Crash the activity, since it saves $1,000,000 for a cost of $300,000

Two days saved avoid $1,000,000 of lost production, which is well above the $300,000 cost, so crashing is economically justified.

Q118 — Answer: C · *People · PE1 Develop a common vision · Agile*

Correct: C. By the team together, linked to the product vision

A mission statement that the team creates together, connected to the product vision, builds ownership and helps the team make consistent decisions.

Q119 — Answer: D · *People · PE8 Plan and manage communication · Agile*

Correct: D. Publish short, user-focused release notes with each release

Short, user-focused release notes tell people what has changed and why it matters to them.

Q120 — Answer: D · *Process · PR2 Develop and manage project scope · Agile*

Correct: D. Write misuse stories alongside the user stories

Misuse (abuser) stories describe what an attacker might try to do, which leads to concrete security requirements and tests.

Q121 — Answer: B · *People · PE8 Plan and manage communication · Agile*

Correct: B. Make the backlog tool the single source of truth and retire the spreadsheets

A single source of truth avoids inconsistent information and reduces the effort of keeping several copies aligned.

Q122 — Answer: C · *People · PE4 Engage stakeholders · Hybrid*

Correct: C. Engage the insurer early to understand its criteria, and include them where they add value

Identifying all stakeholders who can influence or benefit from the project, including insurers, can create additional value.

Q123 — Answer: D · *Business Environment · BE1 Define and establish project governance · Predictive*

Correct: D. Wait for the review, as corporate governance requires, and prepare for it promptly

Governance gates exist to confirm readiness before major commitments. Starting before the review, or working around it, undermines the control that it provides.

Q124 — Answer: B · *Business Environment · BE3 Manage and control changes · Predictive*

Correct: B. It goes through management of change for review and approval before it is made

In major-hazard operations, changes to safety systems, even temporary ones, are reviewed and approved through management of change before they are made.

Q125 — Answer: B · *Business Environment · BE5 Plan and manage risk · Hybrid*

Correct: B. Assign owners, design responses and track each one to verified closure before restart

High-risk recommendations need named owners, effective responses and verified closure before hazardous operation resumes.

Q126 — Answer: C · *Business Environment · BE7 Support organizational change · Hybrid*

Correct: C. Involve operators in designing and testing the screens, and train them on a simulator before go-live

Involving the people affected in the design and giving them safe practice before go-live addresses real concerns and builds confidence.

Q127 — Answer: C · *Process · PR1 Develop an integrated project management plan and plan delivery · Agile*

Correct: C. During refinement and design, through threat modelling

Threat modelling during refinement and design builds security in early, when changes are cheapest.

Q128 — Answer: 12 · *People · PE8 Plan and manage communication · Predictive*

Channels = $n(n - 1)/2$. Solving $n(n - 1)/2 = 66$ gives $n(n - 1) = 132$, so $n = 12$ ($12 \times 11 = 132$).

Q129 — Answer: D · *Process · PR9 Evaluate project status · Hybrid*

Correct: D. Use exception-based reporting, focused on variances, risks and decisions needed

Management by exception focuses the attention of senior stakeholders on what needs their decisions.

Q130 — Answer: D · *People · PE4 Engage stakeholders · Predictive*

Correct: D. An influence/impact grid

Grids such as power/interest and influence/impact help prioritise stakeholders and plan suitable engagement.

Q131 — Answer: D · *Process · PR9 Evaluate project status · Predictive*

Correct: D. Daily, using earned hours against the detailed plan, with quick corrective action

In a short, intense shutdown, daily tracking against the detailed plan allows problems to be spotted and corrected before they become delays.

Q132 — Answer: D · *Process · PR10 Manage project closure · Predictive*

Correct: D. Permits closed, temporary equipment removed and the pre-startup review done

A safe handover requires that work is complete and verified, permits are closed and temporary equipment is removed. Commercial and lessons learned activities can follow.

Q133 — Answer: D · *Process · PR3 Help ensure value-based delivery · Hybrid*

Correct: D. Through outcomes such as fewer crew-related delays, compared with a baseline

Value is shown by measurable outcomes compared with the situation before, not by outputs or effort.

Q134 — Answer: A, D · *People · PE1 Develop a common vision · Hybrid*

Correct: A. Revisiting the goals regularly in joint planning sessions / D. Making goals and progress visible to every location

A common vision is maintained through regular joint reflection and shared visibility, especially when teams work in different places.

Q135 — Answer: B · *Business Environment · BE6 Continuous improvement · Predictive*

Correct: B. Quality circles

Quality circles involve the people who do the work in continuous improvement, using their knowledge of the process.

Q136 — Answer: D · *People · PE4 Engage stakeholders · Hybrid*

Correct: D. Engage the senate early and plan the approval into the schedule

Formal approvals by governance bodies are identified early, and the stakeholders behind them are engaged so that the approvals do not become a bottleneck.

Q137 — Answer: A · *People · PE3 Lead the project team · Agile*

Correct: A. Agree a development plan, such as facilitating some events with coaching and feedback

Servant leaders support people's growth with practical opportunities, coaching and feedback.

Q138 — Answer: A · *Business Environment · BE7 Support organizational change · Hybrid*

Correct: A. Update the procedures, roles and training as part of the change

Organisational change includes aligning formal procedures, roles and training with the new way of working.

Q139 — Answer: A · *Process · PR3 Help ensure value-based delivery · Predictive*

Correct: A. $100,000

PV = FV / (1 + r)^n = 121,000 / 1.1^2 = 121,000 / 1.21 = $100,000.

Q140 — Answer: B · *Process · PR4 Plan and manage resources · Hybrid*

Correct: B. Plan around the teaching timetable and agree their allocation with the department head

Resource availability depends on other commitments. Planning around them and agreeing allocations with the right manager makes plans realistic.

Q141 — Answer: C · *Business Environment · BE4 Remove impediments and manage issues · Agile*

Correct: C. Create anonymised or synthetic test data, agreed with the data protection officer

The impediment is removed in a compliant way: synthetic or anonymised data enables realistic testing without breaching privacy rules.

Q142 — Answer: A · *Process · PR6 Plan and manage finance · Hybrid*

Correct: A. Request approval from the funder before moving the money

Funders' conditions are part of the financial controls. Moving money between categories requires the funder's approval.

Q143 — Answer: B · *Process · PR5 Plan and manage procurement · Hybrid*

Correct: B. Ask the bidder to explain the price, and check that it has understood the scope

An abnormally low bid may reflect a misunderstanding or an unsustainable price. It is clarified before a decision is made.

Q144 — Answer: B · *People · PE5 Align stakeholder expectations · Agile*

Correct: B. Clarify what each MoSCoW category means when priorities are set

MoSCoW only aligns expectations if everyone understands that "should have" items are important but not essential for the release.

Q145 — Answer: A · *Process · PR2 Develop and manage project scope · Predictive*

Correct: A. Add it to the requirements and design, traced back to the HAZOP recommendation

Safety study recommendations become requirements. Tracing them to their source shows that each recommendation has been addressed.

Q146 — Answer: D · *People · PE3 Lead the project team · Agile*

Correct: D. Have a private one-to-one conversation to understand what is going on

A private, caring conversation is the best first step to understand the reasons for a change in behaviour.

Q147 — Answer: A · *Process · PR6 Plan and manage finance · Hybrid*

Correct: A. Add these change management costs to the estimate

Complete estimates include the costs of helping people adopt the change, which are often underestimated or forgotten.

Q148 — Answer: C · *Process · PR8 Plan and manage schedule · Predictive*

Correct: C. At the end of the critical chain, before the project finish date

The project buffer protects the finish date and is placed at the end of the critical chain. Feeding buffers are placed where non-critical chains join it.

Q149 — Answer: A · *Business Environment · BE5 Plan and manage risk · Predictive*

Correct: A. A tornado diagram

A tornado diagram displays the results of sensitivity analysis, with the most influential variables at the top.

Q150 — Answer: C · *Process · PR3 Help ensure value-based delivery · Agile*

Correct: C. Impact mapping

An impact map links a goal to the actors whose behaviour must change, the impacts sought and the deliverables that could cause them.

Q151 — Answer: D · *People · PE1 Develop a common vision · Predictive*

Correct: D. Show how the strategy links to the project's objectives and to their own work

Showing the "line of sight" from strategy to objectives to individual work gives people purpose and helps them prioritise.

Q152 — Answer: D · *Business Environment · BE4 Remove impediments and manage issues · Agile*

Correct: D. When the team cannot remove it and it threatens the sprint goal or delivery

The team tries to remove impediments itself first. Those beyond its control that threaten its goals are escalated promptly.

Q153 — Answer: D · *Business Environment · BE5 Plan and manage risk · Predictive*

Correct: D. A bow-tie analysis

A bow-tie diagram places the hazardous event at the centre, with threats and preventive barriers on the left and consequences and mitigating barriers on the right.

Q154 — Answer: C · *People · PE2 Manage conflicts · Hybrid*

Correct: C. Record both views, refer the dispute to the right forum and keep the rules configurable

The project should not settle labour disputes itself. It documents both views, escalates to the proper forum and keeps the design flexible.

Q155 — Answer: A, D · *People · PE4 Engage stakeholders · Predictive*

Correct: A. Desired engagement levels and strategies / D. The methods and timing of engagement

The stakeholder engagement plan describes the strategies and actions needed to engage stakeholders effectively, including methods and timing.

Q156 — Answer: C · *People · PE2 Manage conflicts · Agile*

Correct: C. Bring both teams together to move security checks earlier and agree shared release criteria

Collaborating on the root cause, which is late feedback, resolves the conflict. Moving checks earlier and agreeing shared criteria serves both teams' interests.

Q157 — Answer: C · *Business Environment · BE3 Manage and control changes · Predictive*

Correct: C. The change goes through the operational change process, with testing and approval

Changes to configuration that affects compliance are controlled through change management, with testing and approval before implementation.

Q158 — Answer: B · *People · PE7 Help ensure knowledge transfer · Agile*

Correct: B. Creating and improving knowledge articles while resolving each ticket

Capturing and improving knowledge as part of resolving each request keeps the knowledge base current and useful.

Q159 — Answer: C · *Process · PR7 Plan and optimize quality of products/deliverables · Predictive*

Correct: C. A pre-startup safety review

A pre-startup safety review verifies that construction meets the design, procedures and training are in place and safety systems work before hazardous materials are introduced.

Q160 — Answer: B · *Process · PR2 Develop and manage project scope · Hybrid*

Correct: B. Measurable reconciliation criteria agreed with the data owners

Objective, measurable acceptance criteria, agreed with those who own the data, make it possible to verify the migration before go-live.

Q161 — Answer: C · *People · PE6 Manage stakeholder expectations · Predictive*

Correct: C. Acknowledge the problem, provide the complete report quickly and agree a reporting schedule

Regulators' trust is maintained by acknowledging failures, correcting them promptly and agreeing reliable arrangements for the future.

Q162 — Answer: C · *People · PE7 Help ensure knowledge transfer · Hybrid*

Correct: C. A short after-action review with the people involved

After-action reviews capture lessons while they are fresh and turn them into improvements for the next event.

Q163 — Answer: A · *Process · PR9 Evaluate project status · Hybrid*

Correct: A. Mean time to detect and mean time to respond

Outcome-focused metrics such as detection and response times show whether the operation is becoming more effective.

Q164 — Answer: A, C · *Business Environment · BE2 Plan and manage project compliance · Hybrid*

Correct: A. Notifying the data protection authority within the legal deadline / C. Informing the affected people when the risk to them is high

Many data protection laws require notification of the authority within a short deadline (often 72 hours) and of the affected people when there is a high risk to them.

Q165 — Answer: D · *People · PE1 Develop a common vision · Hybrid*

Correct: D. An obeya (big room)

An obeya is a lean visual management practice: a room where objectives, progress and problems are displayed so that leaders and teams share the same picture.

Q166 — Answer: D · *Process · PR4 Plan and manage resources · Hybrid*

Correct: D. Use a device lab or a cloud device-testing service

Physical resources such as test devices can be obtained in flexible ways, for example through shared device labs or cloud services.

Q167 — Answer: C · *Business Environment · BE1 Define and establish project governance · Predictive*

Correct: C. A RAID log

A RAID log brings risks, assumptions, issues and dependencies together, which gives governance bodies a compact view of what needs attention.

Q168 — Answer: C · *Process · PR3 Help ensure value-based delivery · Hybrid*

Correct: C. Adopting the tool and adapting processes for the gaps, if the overall value is higher

Value is maximised by comparing options on total cost and benefit. Adapting processes can be cheaper than building every feature.

Q169 — Answer: D · *People · PE2 Manage conflicts · Agile*

Correct: D. Involve a neutral mediator, such as an experienced coach or HR

When facilitation fails, a neutral third party can help the people involved resolve the conflict before it does further harm.

Q170 — Answer: B · *Business Environment · BE3 Manage and control changes · Hybrid*

Correct: B. A tested backout plan

A tested backout (rollback) plan allows the team to restore the previous state quickly if the change causes problems.

Q171 — Answer: A · *People · PE6 Manage stakeholder expectations · Agile*

Correct: A. Explain that forecasts are updated with new data, and show the range and its trend

Agile forecasts improve as data accumulates. Showing the range and how it is narrowing helps stakeholders understand and trust the updates.

Q172 — Answer: B · *Business Environment · BE8 Evaluate external business environment changes · Hybrid*

Correct: B. Engage the community proactively about the upgrade and its safety measures

Social factors, such as public concern, are part of the business environment. Proactive, clear engagement builds trust.

Q173 — Answer: B · *Process · PR2 Develop and manage project scope · Predictive*

Correct: B. Product analysis

Product analysis, including product breakdown, translates a high-level product description into tangible deliverables.

Q174 — Answer: A · *People · PE7 Help ensure knowledge transfer · Hybrid*

Correct: A. Reverse mentoring

In reverse mentoring, junior staff share their expertise with senior colleagues, which transfers knowledge and builds relationships.

Q175 — Answer: C · *Process · PR8 Plan and manage schedule · Predictive*

Correct: C. Very detailed, often down to hours, with shift-level coordination

Shutdowns run around the clock with many trades. Hour-level schedules and shift coordination are needed to control time and cost.

Q176 — Answer: B · *Process · PR10 Manage project closure · Hybrid*

Correct: B. Deposit the data in an approved repository, as the data management plan requires

Funders' data requirements are closure obligations. An approved repository preserves the data and makes it available under the agreed conditions.

Q177 — Answer: A · *Process · PR10 Manage project closure · Predictive*

Correct: A. Demobilise crews as their work is accepted, checking tools, permits and badges

Releasing resources as their work is accepted controls cost, while administrative checks close out access and equipment properly.

Q178 — Answer: B · *People · PE2 Manage conflicts · Predictive*

Correct: B. Remove her from the evaluation and record the disclosure

A disclosed conflict of interest is managed by removing the person from the decision and documenting it. The bidder is not penalised for the panel member's relationship.

Q179 — Answer: D · *Business Environment · BE8 Evaluate external business environment changes · Predictive*

Correct: D. Plan an upgrade or replacement, and assess the risks until it is complete

Technology obsolescence is a predictable external change. Planning ahead avoids running critical systems without security updates.

Q180 — Answer: B · *Process · PR7 Plan and optimize quality of products/deliverables · Predictive*

Correct: B. The process cannot consistently meet the specification limits

A Cpk below 1.0 means that the process spread, or its position, does not fit within the specification limits, so defects are expected. A Cpk of 1.33 or more is often required.

Score Analysis

Count your correct answers in each domain and task, then compute the percentage. A multiple-response or matching question counts as correct only if **every** part is right (as on the real exam).

Domain	Questions	Your correct	%
People	59		
Process	74		
Business Environment	47		
Total	**180**		

Task	Questions (numbers)
PE1 — Develop a common vision	7 (1, 105, 116, 118, 134, 151, 165)
PE2 — Manage conflicts	8 (16, 59, 63, 65, 154, 156, 169, 178)
PE3 — Lead the project team	9 (19, 44, 49, 50, 57, 74, 109, 137, 146)
PE4 — Engage stakeholders	8 (34, 66, 84, 95, 122, 130, 136, 155)
PE5 — Align stakeholder expectations	7 (17, 46, 75, 99, 101, 110, 144)
PE6 — Manage stakeholder expectations	7 (8, 10, 29, 56, 62, 161, 171)
PE7 — Help ensure knowledge transfer	6 (9, 13, 86, 158, 162, 174)
PE8 — Plan and manage communication	7 (12, 25, 27, 39, 119, 121, 128)
PR1 — Develop an integrated project management plan and plan delivery	8 (2, 3, 33, 47, 73, 102, 113, 127)
PR2 — Develop and manage project scope	8 (35, 45, 51, 111, 120, 145, 160, 173)
PR3 — Help ensure value-based delivery	8 (14, 18, 78, 90, 133, 139, 150, 168)
PR4 — Plan and manage resources	7 (4, 31, 54, 61, 64, 140, 166)
PR5 — Plan and manage procurement	7 (20, 22, 28, 68, 106, 108, 143)
PR6 — Plan and manage finance	8 (55, 58, 76, 79, 93, 117, 142, 147)
PR7 — Plan and optimize quality of products/deliverables	7 (7, 30, 38, 81, 92, 159, 180)
PR8 — Plan and manage schedule	8 (67, 82, 85, 97, 103, 107, 148, 175)
PR9 — Evaluate project status	7 (15, 24, 32, 100, 129, 131, 163)
PR10 — Manage project closure	6 (6, 21, 88, 132, 176, 177)
BE1 — Define and establish project governance	6 (40, 41, 60, 69, 123, 167)
BE2 — Plan and manage project compliance	6 (5, 26, 37, 43, 80, 164)
BE3 — Manage and control changes	6 (11, 71, 114, 124, 157, 170)
BE4 — Remove impediments and manage issues	6 (36, 48, 53, 112, 141, 152)
BE5 — Plan and manage risk	7 (83, 87, 89, 96, 125, 149, 153)

Task	Questions (numbers)
BE6 — Continuous improvement	5 (70, 77, 91, 98, 135)
BE7 — Support organizational change	6 (23, 52, 94, 104, 126, 138)
BE8 — Evaluate external business environment changes	5 (42, 72, 115, 172, 179)

How to read your result (self-assessment guide — PMI does not publish a fixed pass mark; the real exam reports *Above Target / Target / Below Target / Needs Improvement* per domain):

Your %	Indicative rating	What to do
80% or more	Above Target	You are ready. Keep practising timing.
70–79%	Target	Likely pass. Review every missed question and its domain.
60–69%	Below Target	Re-study the weak domains in the Study Guide before the next exam.
under 60%	Needs Improvement	Go back to the Study Guide chapters and the Question Bank first.